

SUPPORTING INFORMATION

Development of degraders of cyclin-dependent kinases 4 and 6 based on rational drug design

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1. Figure S1. Western blotting analysis of CDK4/6 protein in Jurkat.

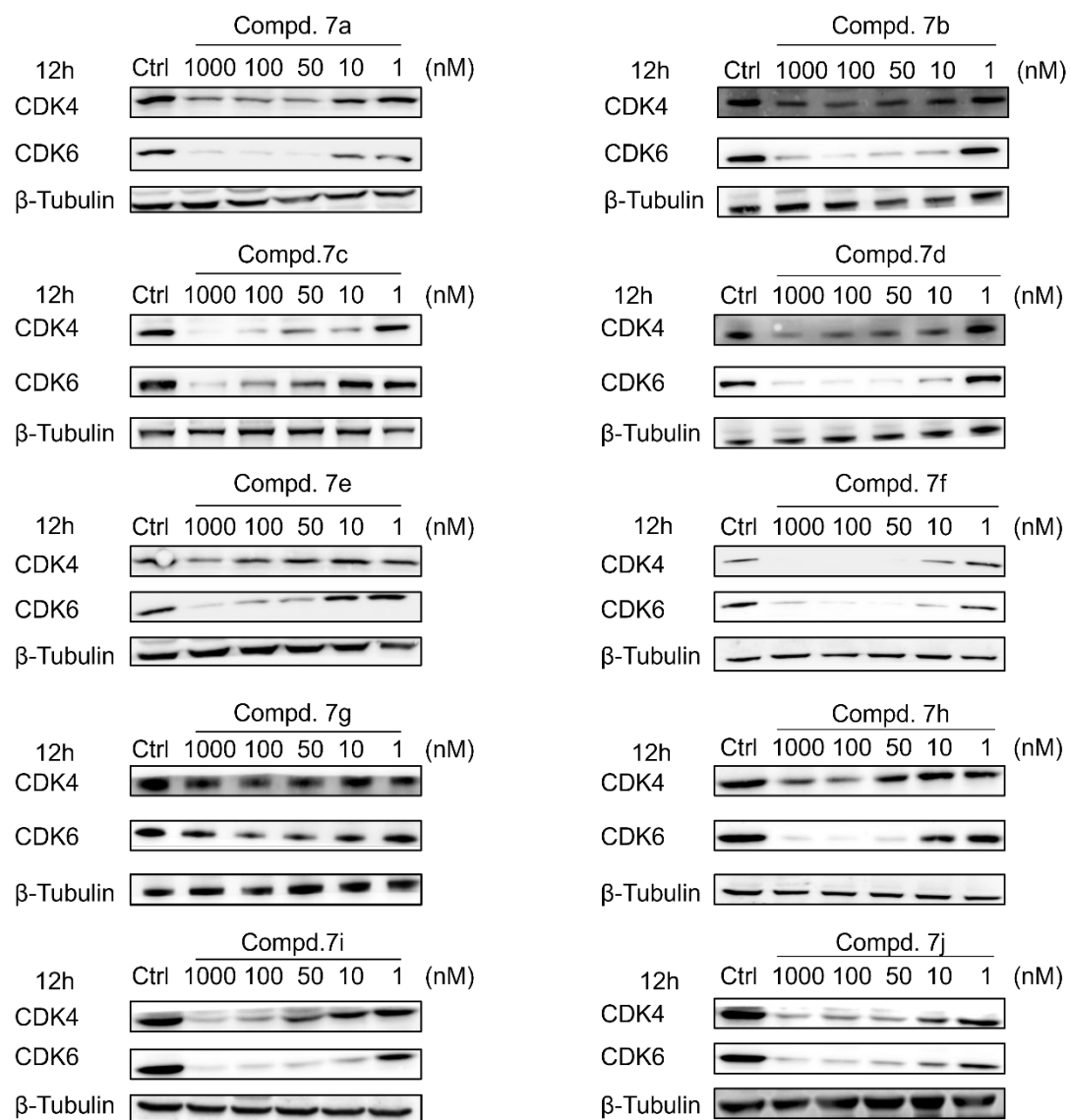
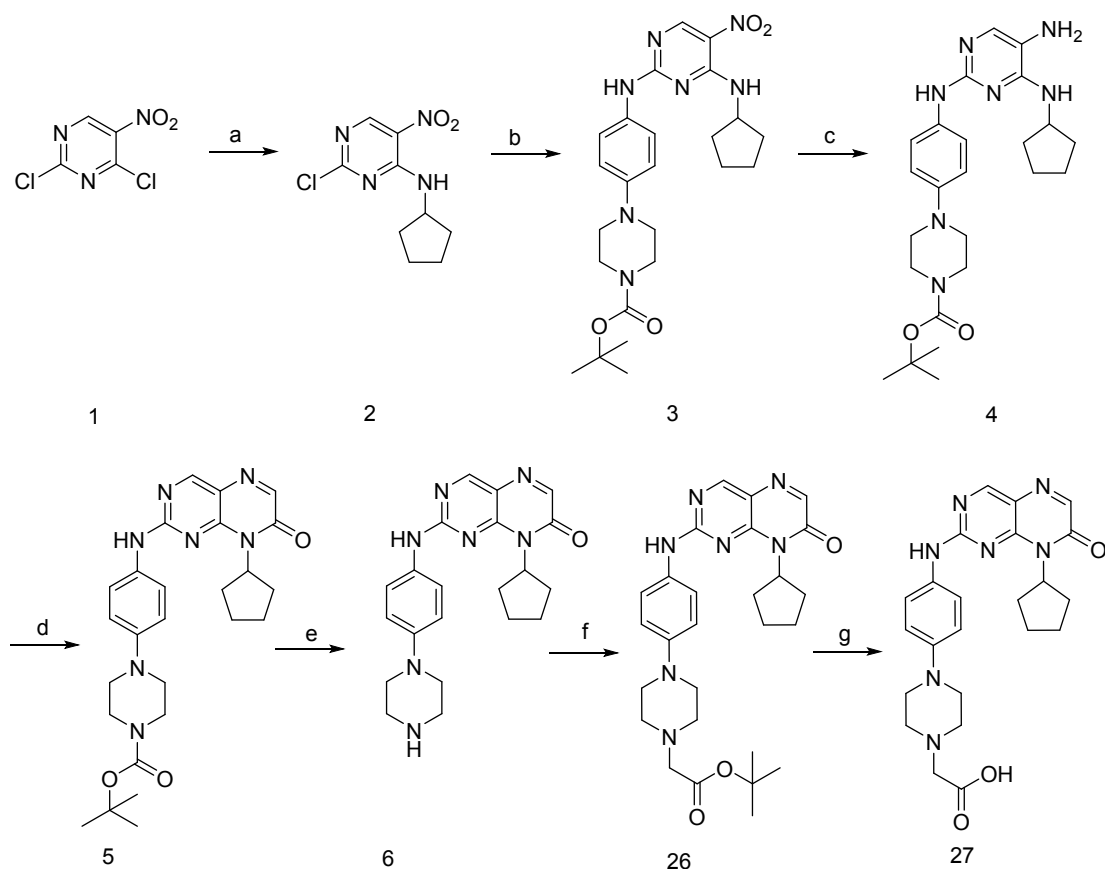


Fig. S1. **7a-7j** induced concentration dependent degradation of CDK4/6 protein in Jurkat cells.

2. Experimental Section—Chemistry

2.1 Scheme S1. Synthesis of Key Intermediate **6** and **27**^a



^aReagents and conditions: (a) Cyclopentylamine, NaHCO₃, DCM, r.t., 6 h, 93.5%; (b) *N*, *N*-dimethylaniline, IPA, THF, 80 °C, 5 h, 83.8%; (c) 10%Pd/C, H₂, MeOH, 40 °C, 3 h, 69.1%; (d) Ethyl glyoxalate solution, CH₃COOH, EtOH, 100 °C, 45 min, 18.5%; (e) TFA, DCM, r.t., 1 h, 66.0%. (f) K₂CO₃, MeCN, 80 °C, 5 h, 54.2%; (g) TFA, DCM, r.t., 1 h, 93.6%.

2-chloro-N-cyclopentyl-5-nitropyrimidin-4-amine (2). To the solution of 2,4-dichloro-5-nitropyrimidine (50.00 g, 259.00 mmol) in dichloromethane (200 mL) were added cyclopentylamine (26.40 g, 311.00 mmol) and NaHCO₃ (43.50 g, 518.00 mmol). The reaction was stirred at room temperature for 6 h, then washed with water and the organic layer was dried with anhydrous sodium sulfate, and concentrated under reduced pressure to remove the solvent to obtain compound **2** (58.50 g, 93.5%) as yellow solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 9.00 (s, 1H), 8.61 (d, *J* = 7.6 Hz, 1H), 4.53 – 4.44 (m, 1H), 2.00 – 1.95 (m, 2H), 1.75 – 1.69 (m, 2H), 1.67 – 1.62 (m, 2H), 1.60 – 1.56 (m, 2H). LC-MS (ESI): *m/z*: 243.30 [M+H]⁺.

tert-butyl 4-(4-((4-(cyclopentylamino)-5-nitropyrimidin-2-yl)amino)phenyl)piperazine-1-carboxylate (3). 2-chloro-*N*-cyclopentyl-5-nitropyrimidin-4-amine (3.00 g, 0.01 mol), *N,N*-dimethylaniline (1.80 g, 0.01 mol) was dissolved in THF (20 mL), then the solution of tert-butyl 4-(4-aminophenyl)piperazine-1-carboxylate (3.77 g, 0.01 mol) in IPA (30 mL) was added. The reaction was stirred at 80 °C for 5 h, then cooled to room temperature, and the pH was adjusted with acetic acid, after adding the right amount of water, we get compound 3 (5.00 g, 83.8%) as yellow solid.

tert-butyl 4-(4-((5-amino-4-(cyclopentylamino)pyrimidin-2-yl)amino)phenyl)piperazine-1-carboxylate (4). To the solution of tert-butyl 4-(4-((4-(cyclopentylamino)-5-nitropyrimidin-2-yl)amino)phenyl)piperazine-1-carboxylate (2.00 g, 4.14 mmol) in methanol (30 mL) were added 10% palladium carbon, then the three-necked flask was degassed and charged with H₂, the reaction was stirred at 40 °C for 3 h, after the reaction was completed, filtered by diatomaceous earth, the solvent was removed and then give compound 4 (1.30 g, 69.1%) as gray solid. ¹H NMR (400 MHz, CDCl₃) δ 7.55 (s, 1H), 7.50 (d, *J* = 8.8 Hz, 2H), 6.89 (d, *J* = 8.7 Hz, 2H), 6.71 (s, 1H), 5.16 (d, *J* = 6.9 Hz, 1H), 4.40 – 4.30 (m, 1H), 3.57 (t, *J* = 5.1 Hz, 4H), 3.04 (t, *J* = 5.1 Hz, 4H), 2.52 (s, 2H), 2.13 – 2.06 (m, 2H), 1.76–1.71 (m, 2H), 1.67–1.62 (m, 2H), 1.55 – 1.50 (m, 2H), 1.48 (s, 9H). LC-MS (ESI): *m/z*: 452.20 [M-H]⁻.

tert-butyl 4-(4-((8-cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazine-1-carboxylate (5). To the solution of tert-butyl 4-(4-((5-amino-4-(cyclopentylamino)pyrimidin-2-yl)amino)phenyl)piperazine-1-carboxylate (1.00 g, 2.20 mmol) in ethanol (4 mL) were added ethyl glyoxylate (0.22 g, 2.20 mmol) and acetic acid (1 mL), the resulting mixture was stirred at 100 °C, 50 w, 150 psi for 45 min, after the reaction was completed, the pH was adjusted with ammonia, and then washed with water and the organic phase was concentrated to dryness. Flash column chromatography (DCM/MeOH = 50/1, v/v) to give compound 5 (0.20 g, 18.5%) as yellow solid. ¹H NMR (600 MHz, CDCl₃) δ 7.98 (s, 1H), 7.93 (s, 1H), 7.53 (d, *J* = 8.9 Hz, 2H), 7.19 (s, 1H), 6.91 (d, *J* = 9.0 Hz, 2H), 4.40 – 4.37 (m,

1H), 3.58 (t, $J = 5.2$ Hz, 4H), 3.08 (t, $J = 5.2$ Hz, 4H), 2.13 – 2.06 (m, 2H), 1.79 – 1.73 (m, 2H), 1.67 – 1.61 (m, 2H), 1.59 – 1.52 (m, 2H), 1.48 (s, 9H). LC-MS (ESI): m/z : 492.30 $[M+H]^+$.

8-cyclopentyl-2-((4-(piperazin-1-yl)phenyl)amino)pteridin-7(8H)-one (6). To the solution of tert-butyl tert-butyl 4-(4-((8-cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazine-1-carboxylate (200 mg, 0.41 mmol) in dichloromethane (4 mL) were added TFA (1 mL), the reaction was stirred at room temperature for 1 h, after the reaction was completed, the solvent was removed and ammonia was added and ground to obtain compound 6 (105 mg, 66.0%) as yellow solid. ^1H NMR (400 MHz, $\text{DMSO-}d_6$) δ 10.00 (s, 1H), 8.77 (s, 2H), 7.84 (s, 1H), 7.55 (d, $J = 8.6$ Hz, 2H), 7.00 (d, $J = 8.8$ Hz, 2H), 5.71 – 5.55 (m, 1H), 3.29 (t, 4H), 3.25 (t, 4H), 2.26 – 2.18 (m, 2H), 1.92 – 1.83 (m, 2H), 1.81 – 1.74 (m, 2H), 1.60 – 1.55 (m, 2H), 1.22 (s, 1H). LC-MS (ESI): m/z : 392.20 $[M+H]^+$.

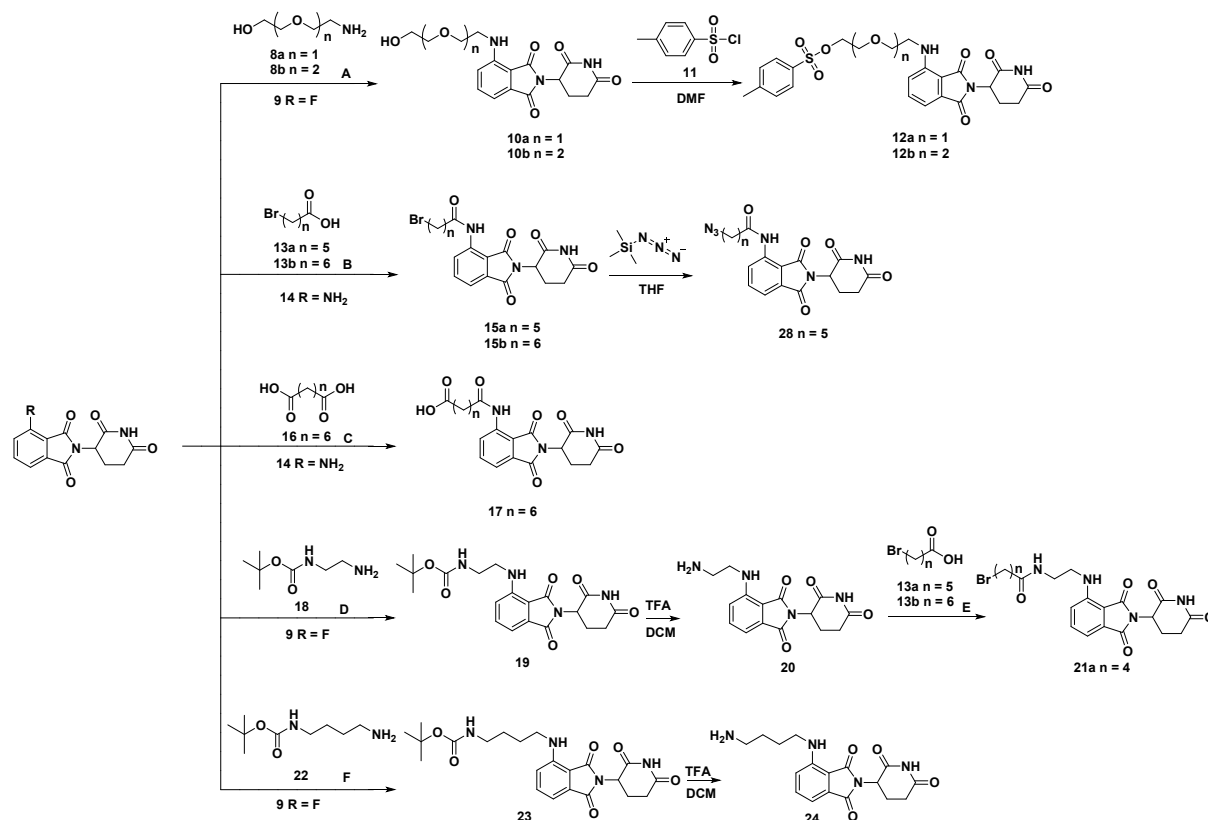
2-(4-(4-((8-cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)acetate (26). To the solution of 8-cyclopentyl-2-((4-(piperazin-1-yl)phenyl)amino)pteridin-7(8H)-one (6) (50.00 mg, 0.13 mmol) in acetonitrile (1 mL) were added tert-butyl bromoacetate (25) (25.00 mg, 0.13 mmol) and K_2CO_3 (54.00 mg, 0.39 mmol). The reaction was stirred at 80°C for 5 h, then washed with water and the organic phase was concentrated to dryness. Flash column chromatography (DCM/MeOH = 150/1, v/v) to give compound (26) (35.00 mg, 54.2%) as yellow solid. ^1H NMR (400 MHz, $\text{DMSO-}d_6$) δ 9.94 (s, 1H), 8.76 (s, 1H), 7.83 (s, 1H), 7.50 (d, $J = 8.7$ Hz, 2H), 6.93 (d, $J = 9.0$ Hz, 2H), 5.73 – 5.54 (m, 1H), 3.15 (s, 2H), 3.09 (t, $J = 5.0$ Hz, 4H), 2.64 (t, $J = 5.0$ Hz, 4H), 2.25 – 2.18 (m, 2H), 1.97 – 1.84 (m, 2H), 1.81 – 1.74 (m, 2H), 1.61 – 1.55 (m, 2H), 1.42 (s, 9H). LC-MS (ESI): m/z : 506.29 $[M+H]^+$.

2-(4-(4-((8-cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)acetic acid (27). To the solution of tert-butyl 2-(4-(4-((8-cyclopentyl-7-oxo-7,8-dihydropteridin-2-yl)amino)phenyl)piperazin-1-yl)acetate (26) (30.00 mg, 0.06 mmol) in dichloromethane (2 mL) were added TFA (0.5 mL) at room temperature. The reaction was stirred overnight, and then the solvent was removed

and the product was directly proceeded to the next step without purification.

8-cyclopentyl-2-((4-(4-(prop-2-yn-1-yl)piperazin-1-yl)phenyl)amino)pteridin-7(8H)-one (29). To the solution of 8-cyclopentyl-2-((4-(piperazin-1-yl)phenyl)amino)pteridin-7(8H)-one (**6**) (50.00 mg, 0.13 mmol) in DMF (1 mL) were added K₂CO₃ (22.00 mg, 0.16 mmol), TBAB (4.00 mg, 0.01 mmol) and 3-bromopropyne (17.00 mg, 0.14 mmol). The reaction was stirred at 90°C for 3 h, and then washed with water and the organic phase was concentrated to dryness. Flash column chromatography (DCM/MeOH = 100/1, v/v) to give compound (**29**) (40.00 mg, 72.7%) as yellow solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 9.94 (s, 1H), 8.76 (s, 1H), 7.83 (s, 1H), 7.51 (d, *J* = 8.6 Hz, 2H), 6.94 (d, *J* = 9.0 Hz, 2H), 5.70 – 5.57 (m, 1H), 3.19 (s, 1H), 3.12 (t, *J* = 5.3 Hz, 4H), 2.61 (t, 4H), 2.26 – 2.18 (m, 2H), 1.93 – 1.83 (m, 2H), 1.82 – 1.75 (m, 2H), 1.62 – 1.56 (m, 2H).

2.2 Scheme S2. Synthesis of key linkers with E3 ligands^a



^aReagents and conditions: (A) DMF, DIPEA, 90 °C, 4 h; (B) DMF, SOCl₂, 80 °C, 2 h; (C,) DCM, HOBT, EDCI, r.t., 12 h; (D) DMF, DIPEA, 90 °C, 4 h; (E) DCM, HATU, DIPEA, r.t., 12 h; (F) DMF, DIPEA, 90 °C, 4 h.

2-(2,6-dioxopiperidin-3-yl)-4-((2-(2-hydroxyethoxy)ethyl)amino)isoindoline-1,3-dione (10a). To the solution of 2-(2,6-dioxopiperidin-3-yl)-4-fluoroisoindoline-1,3-dione (**9**) (500.00 mg, 1.81 mmol) in DMF (5 mL) were added diethylene glycolamine (**8a**) (228.00 mg, 2.17 mmol) and *N,N*-diisopropylethylamine (1 mL). The reaction was stirred at 90°C for 5 h, and then washed with water and saturated salt water and the organic phase was concentrated to dryness, the product was chromatographed on a 200-300 mesh silica gel column (DCM/MeOH = 100/1, v/v) to give compound (**10a**) (320.00 mg, 48.9%) as bright yellow solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.10 (s, 1H), 7.59 (dd, *J* = 8.6, 7.1 Hz, 1H), 7.15 (d, *J* = 8.6 Hz, 1H), 7.04 (d, *J* = 7.0 Hz, 1H), 6.61 (t, *J* = 5.9 Hz, 1H), 5.06 (dd, *J* = 12.9, 5.4 Hz, 1H), 4.62 (t, *J* = 5.3 Hz, 1H), 3.62 (t, *J* = 5.5 Hz, 2H), 3.53 – 3.49 (m, 2H), 3.48 – 3.44 (m, 4H), 2.64 – 2.51 (m, 2H).

2-(2,6-dioxopiperidin-3-yl)-4-((2-(2-(2-hydroxyethoxy)ethoxy)ethyl)amino)isoindoline-1,3-dione (10b). General procedure for synthesis of compound **10b**. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.10 (s, 1H), 7.58 (t, 1H), 7.15 (d, *J* = 8.6 Hz, 1H), 7.04 (d, *J* = 7.1 Hz, 1H), 6.61 (t, *J* = 5.8 Hz, 1H), 5.05 (dd, *J* = 13.0, 5.4 Hz, 1H), 4.55 (t, *J* = 5.4 Hz, 1H), 3.62 (t, *J* = 5.5 Hz, 2H), 3.58 – 3.55 (m, 2H), 3.55 – 3.52 (m, 2H), 3.50 – 3.45 (m, 4H), 3.41 (t, *J* = 4.8 Hz, 2H), 2.64 – 2.51 (m, 2H), 2.08 – 1.97 (m, 2H).

6-bromo-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)hexanamide (15a). To the solution of 6-bromohexanoic acid (**13a**) (1.00 g, 5.13 mmol) in sulfoxide chloride (10 mL) were added one drop of DMF. The reaction was stirred at 80°C for 3 h, and then the solvent was removed and *N,N*-diisopropylethylamine (2 mL), THF (15 mL) and pomalidomide (**14**) were added, the resulting mixture was stirred at 50°C for 6 h. After the reaction, the product was washed with a large amount of aqueous hydrochloric acid, the organic phase was concentrated to dryness, the product was chromatographed on a 200-300 mesh silica gel column (DCM/MeOH = 150/1, v/v) to give compound (**15a**) (580.00 mg, 25.2%) as white solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.15 (s, 1H), 9.73 (s, 1H), 8.45 (d, *J* = 8.4 Hz, 1H), 7.83 (t, *J* = 7.9 Hz, 1H), 7.61 (d, *J* = 7.3 Hz, 1H), 5.14 (dd, *J* = 12.7, 5.5 Hz, 1H), 2.31 – 2.25 (m,

2H), 2.20 – 2.11 (m, 2H), 1.86 – 1.78 (m, 2H), 1.71 – 1.69 (m, 2H), 1.63 – 1.60 (m, 2H), 1.51 – 1.48 (m, 2H), 1.39 – 1.36 (m, 2H).

7-bromo-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)heptanamide (15b). General procedure for synthesis of compound **15b**. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.15 (s, 1H), 9.71 (s, 1H), 8.46 (d, *J* = 8.4 Hz, 1H), 7.83 (t, *J* = 7.9 Hz, 1H), 7.61 (d, *J* = 7.3 Hz, 1H), 5.14 (dd, *J* = 12.8, 5.4 Hz, 1H), 3.58 (dt, *J* = 40.2, 6.6 Hz, 2H), 2.67 – 2.51 (m, 2H), 2.46 (t, *J* = 7.4 Hz, 2H), 1.85 – 1.68 (m, 2H), 1.67 – 1.57 (m, 2H), 1.47 – 1.30 (m, 4H).

8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-8-oxooctanoic acid (17). ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.98 (s, 1H), 11.15 (s, 1H), 9.70 (s, 1H), 8.46 (d, *J* = 8.4 Hz, 1H), 7.83 (t, *J* = 7.9 Hz, 1H), 7.61 (d, *J* = 7.3 Hz, 1H), 5.14 (dd, *J* = 12.8, 5.4 Hz, 1H), 2.66 – 2.52 (m, 2H), 2.46 (t, *J* = 7.5 Hz, 2H), 2.20 (t, *J* = 7.3 Hz, 2H), 1.67 – 1.58 (m, 2H), 1.53 – 1.46 (m, 2H), 1.38 – 1.29 (m, 4H), 1.28 – 1.17 (m, 2H).

(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)carbamate (19). To the solution of 2-(2,6-dioxopiperidin-3-yl)-4-fluoroisindoline-1,3-dione (**9**) (500.00 mg, 1.81 mmol) in DMF (5 mL) were added tert-butyl (2-aminoethyl)carbamate (**18**) (348.00 mg, 2.17 mmol) and *N,N*-diisopropylethylamine (2 mL). The reaction was stirred at 90°C for 5 h and then washed with water and the organic layer was dried with anhydrous sodium sulfate, and concentrated to remove the solvent. Flash column chromatography (DCM/MeOH = 150/1, v/v) to give compound (**19**) (142.00 mg, 18.8%) as bright yellow solid. ¹H NMR (400 MHz, DMSO-*d*₆) δ 11.10 (s, 1H), 7.58 (t, 1H), 7.14 (d, *J* = 8.6 Hz, 1H), 7.06 – 6.99 (m, 2H), 6.72 (t, *J* = 6.2 Hz, 1H), 5.05 (dd, *J* = 12.8, 5.4 Hz, 1H), 3.39 – 3.36 (m, 2H), 3.11 (q, 2H), 2.62 – 2.51 (m, 2H), 2.10 – 1.90 (m, 2H), 1.36 (s, 9H). General procedure for synthesis of compound **23**.

4-((2-aminoethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isindoline-1,3-dione (20). To the solution of tert-butyl (2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)carbamate (**19**) (140.00 mg, 0.34 mmol) in dichloromethane (2 mL) were added TFA (0.5 mL). The reaction was stirred at room

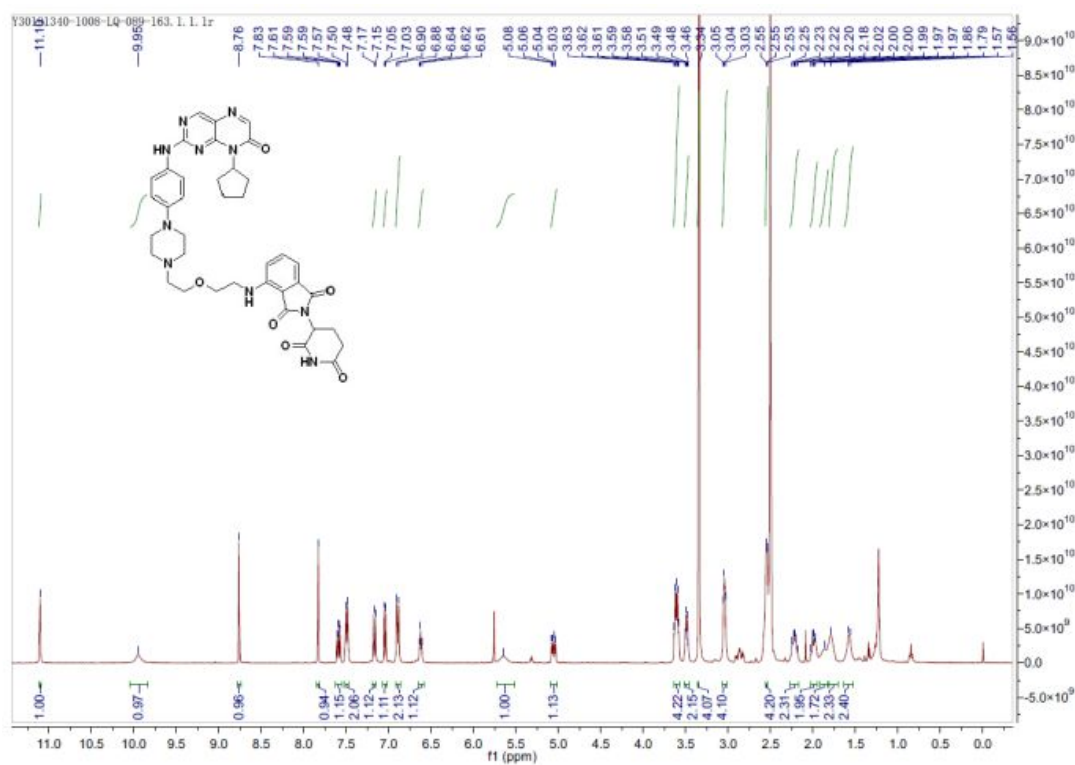
temperature for 1 h, after the reaction was completed, the solvent was removed to give compound (20) (104.00 mg, 98.1%) as yellow oily liquid. **General procedure for synthesis of compound 24.**

5-bromo-N-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)pentanamide (21a). To the solution of 4-((2-aminoethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (20) (100.00 mg, 0.32 mmol) in dichloromethane (2 mL) were added, 5-bromovaleric acid (63.00 mg, 0.35 mmol), HATU (144.00 mg, 0.38 mmol) and DIPEA (0.5 mL). The reaction was stirred at room temperature for 5 h, then washed with water and the organic phase was concentrated to dryness. Flash column chromatography (DCM/MeOH = 60/1, v/v) to give compound (21a) (85.00 mg, 56.1%) as yellow solid. ¹H NMR (400 MHz, DMSO-d₆) δ 11.10 (s, 1H), 8.06 (t, J = 5.6 Hz, 1H), 7.61 – 7.56 (m, 1H), 7.18 – 7.16 (m, 1H), 7.03 (d, J = 7.0 Hz, 1H), 6.72 (t, J = 6.0 Hz, 1H), 5.05 (dd, J = 12.8, 5.4 Hz, 1H), 3.49 (t, J = 6.6 Hz, 2H), 3.27 – 3.22 (m, 2H), 2.62 – 2.54 (m, 2H), 2.11 – 2.06 (m, 2H), 2.06 – 1.97 (m, 2H), 1.77 – 1.73 (m, 2H), 1.63 – 1.57 (m, 2H), 1.29 – 1.25 (m, 2H).

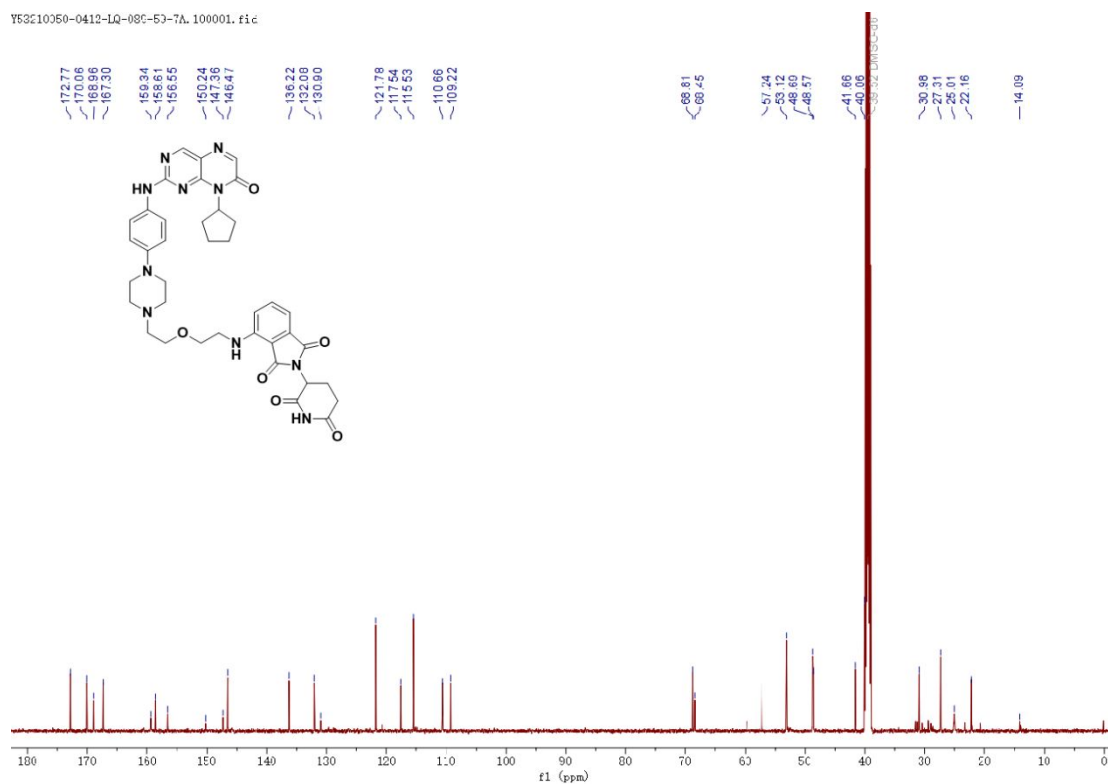
6-azido-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)hexanamide (28). To the solution of 6-bromo-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)hexanamide (15a) (200 mg, 0.44 mmol) in tetrabutylammonium fluoride tetrahydrofuran solution (2 mL) were added azidotrimethylsilane (1 mL) at room temperature. The reaction was stirred overnight, and then the solvent was removed and the product was directly proceeded to the next step without purification.

The spectra of ^1H NMR, ^{13}C NMR, HRMS (ESI) and HPLC of compounds.

1. Spectra of compound 7a (^1H NMR, ^{13}C NMR HRMS, HPLC)



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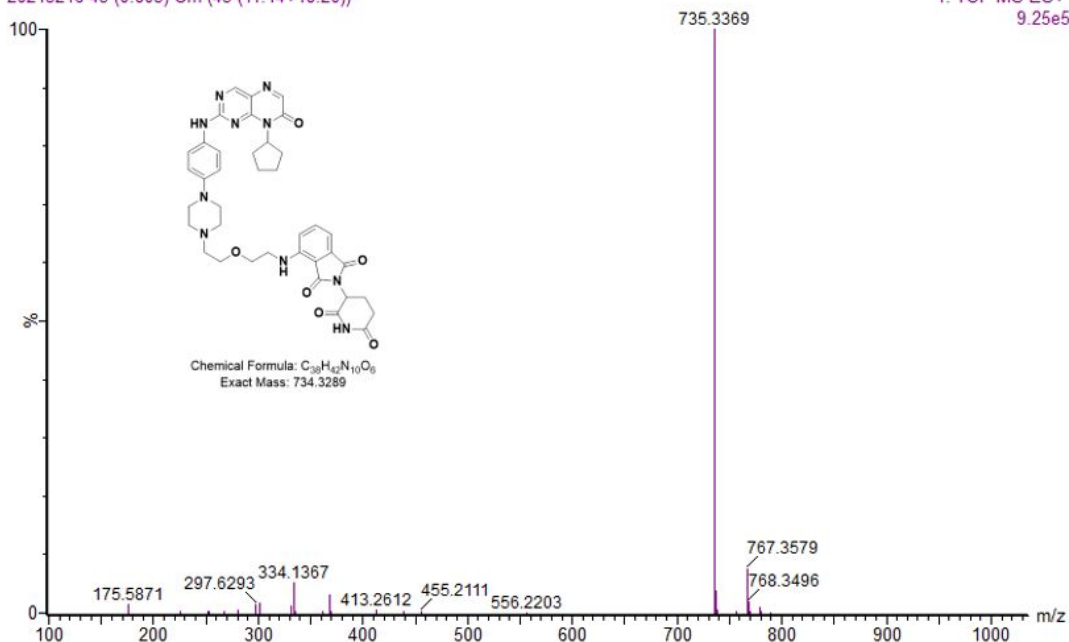
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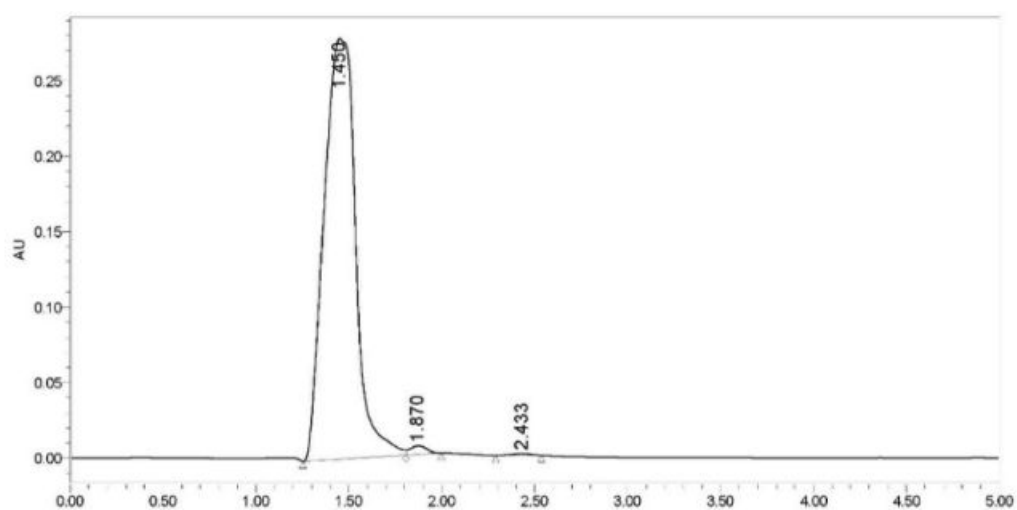
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20213216 43 (0.803) Cm (43-(11:14+18:20))

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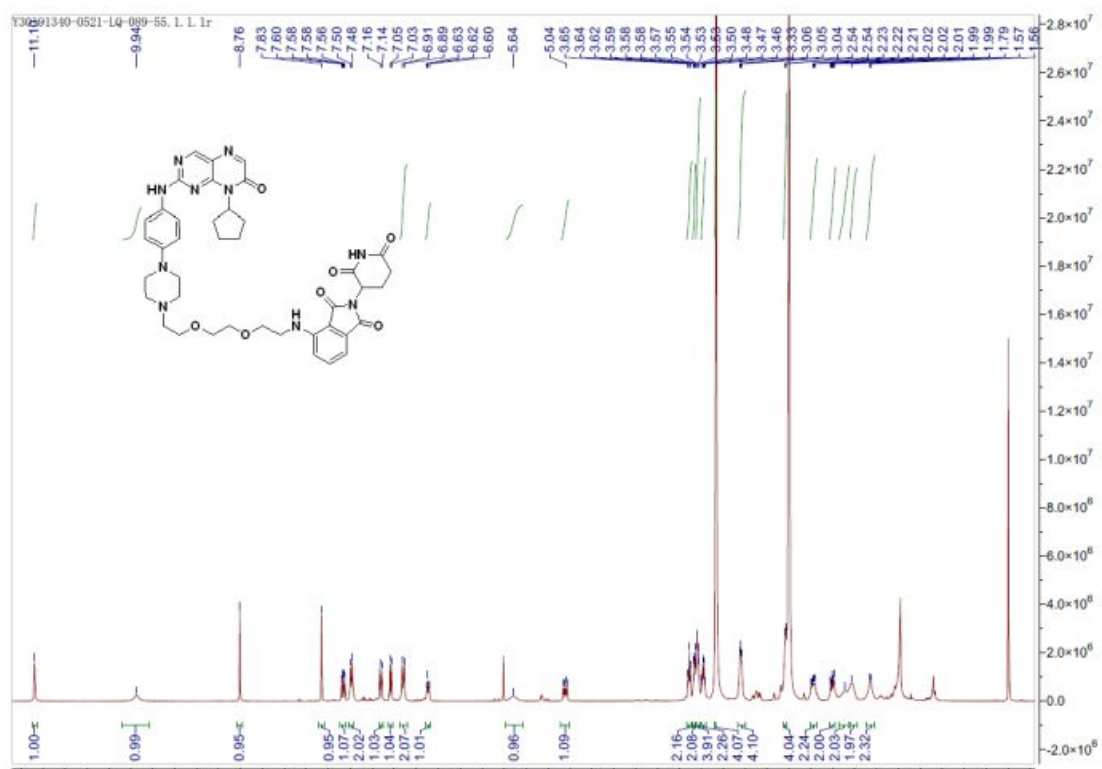
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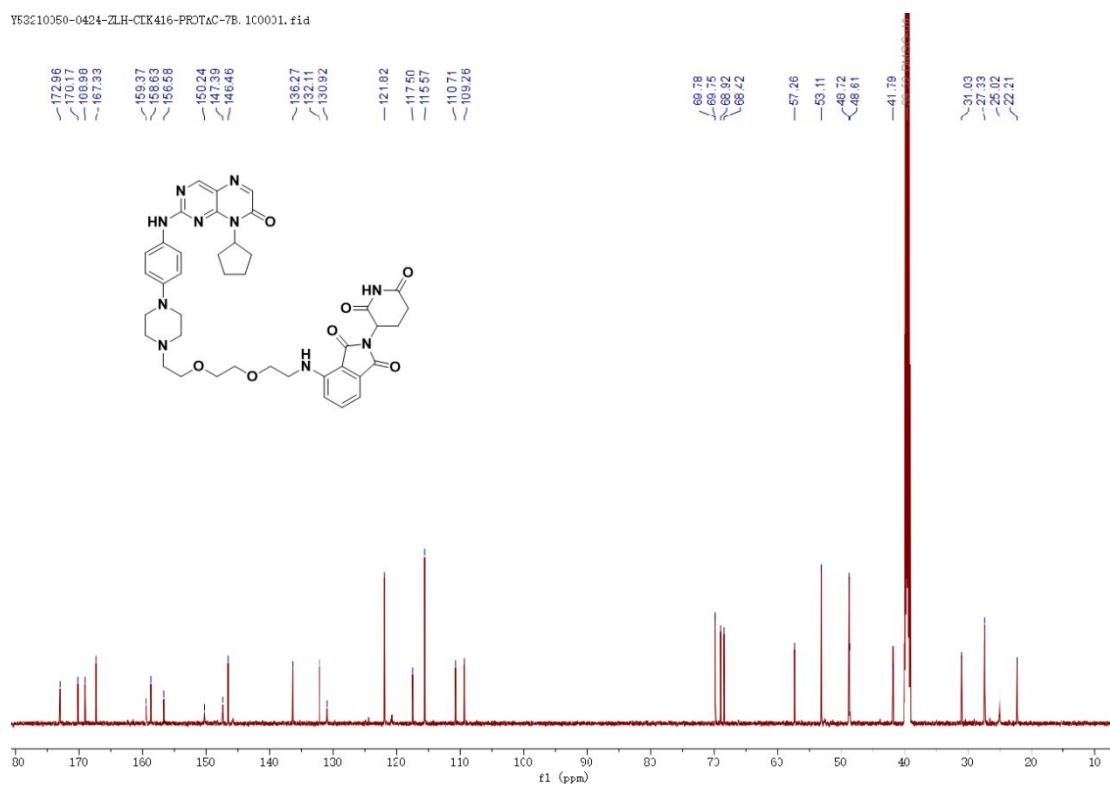


No.	Retention (min)	Area	Area%
1	1.450	3293785	98.65
2	1.870	36105	1.08
3	2.433	9086	0.27

2. Spectra of compound 7b (^1H NMR, ^{13}C NMR HRMS, HPLC)



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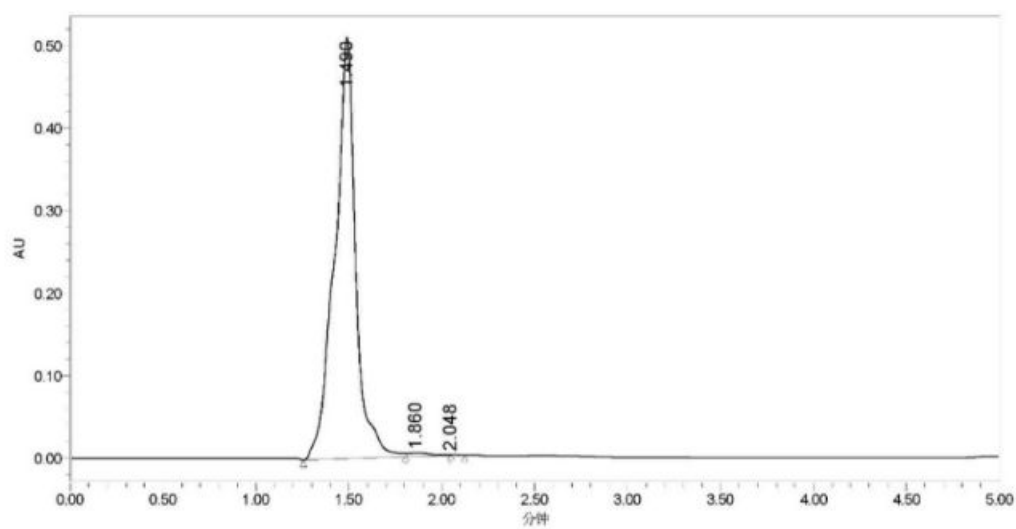
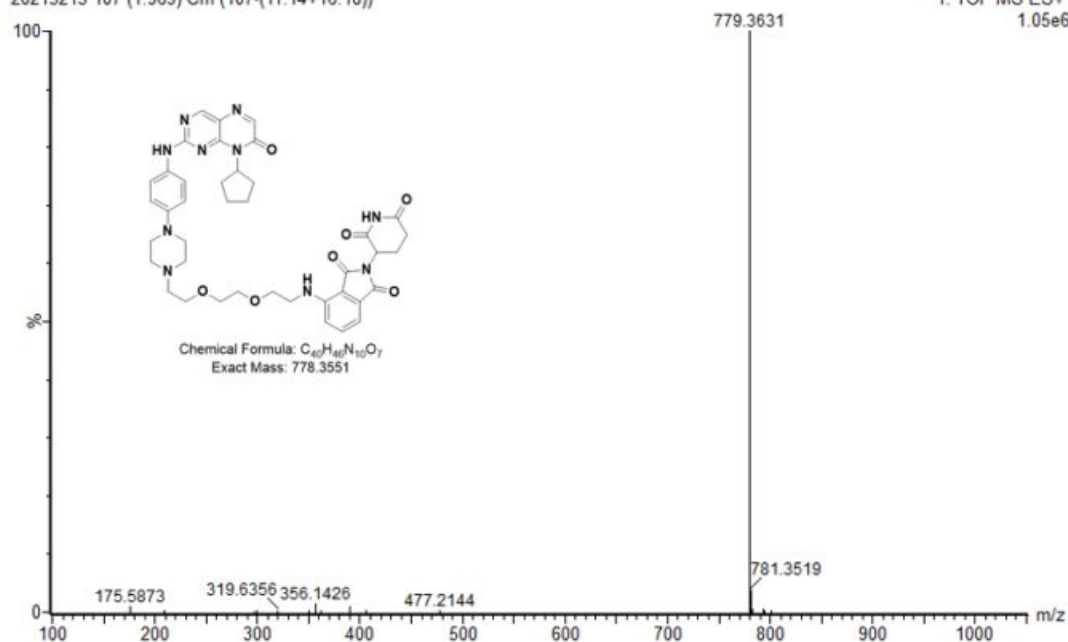


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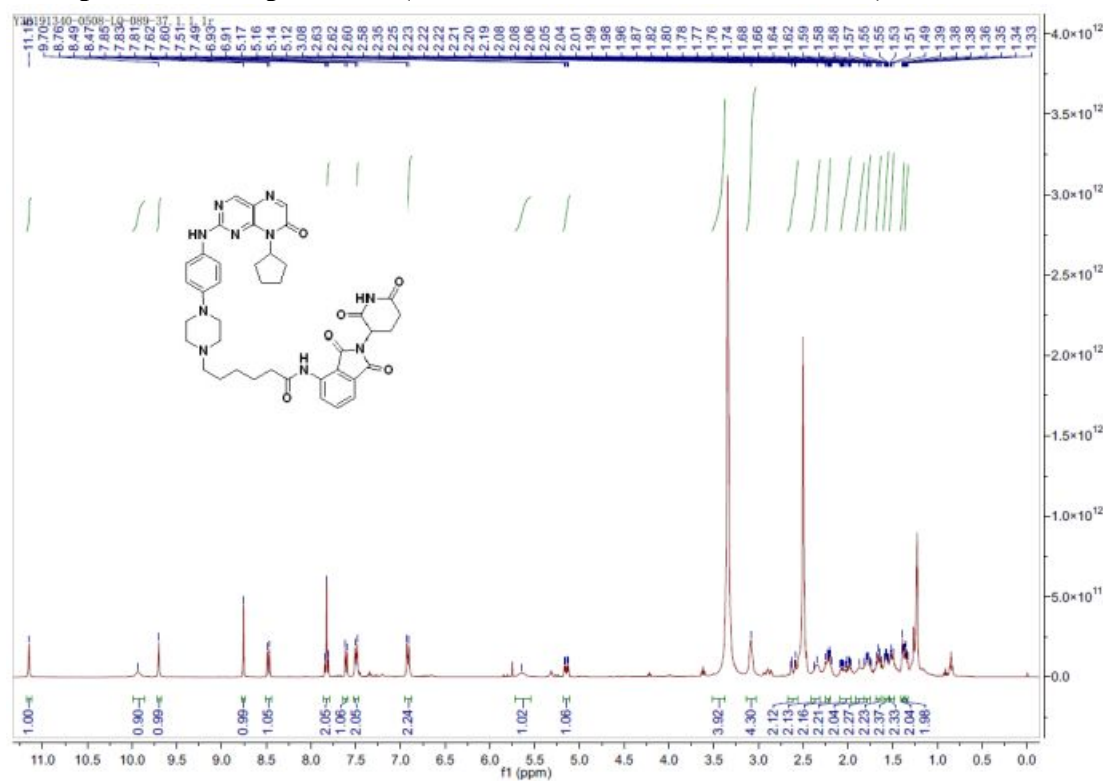
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20213215 107 (1.969) Cm (107-(11:14+16:18))

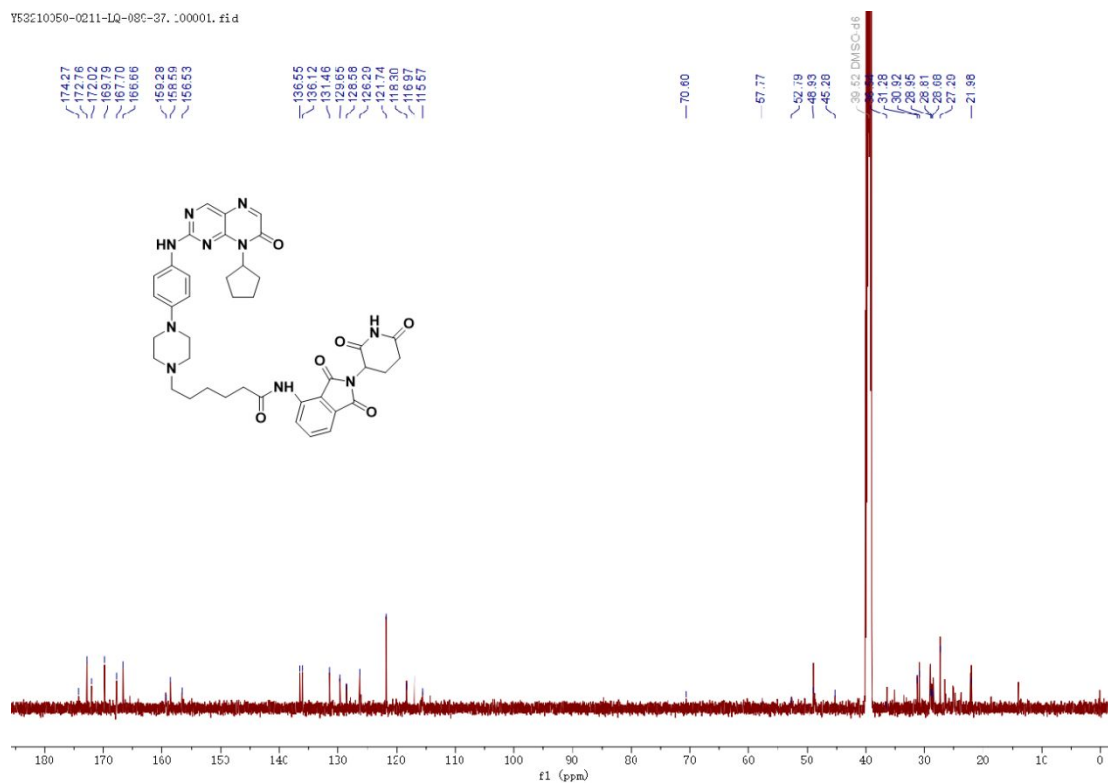
1: TOF MS ES+
1.05e6

No.	Retention (min)	Area	Area%
1	1.490	3991255	98.94
2	1.860	40809	1.01
3	2.048	2135	0.05

3. Spectra of compound 7c (¹H NMR, ¹³C NMR HRMS, HPLC)



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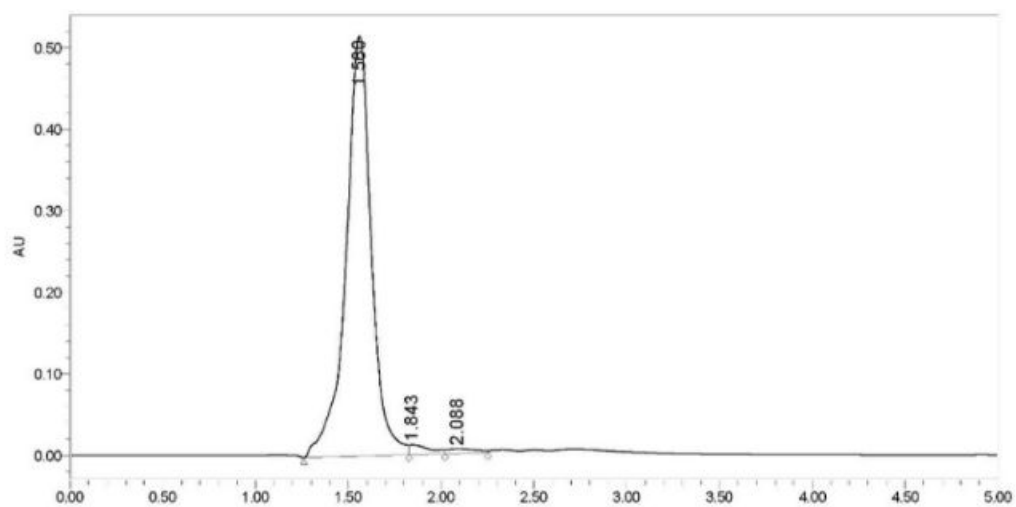
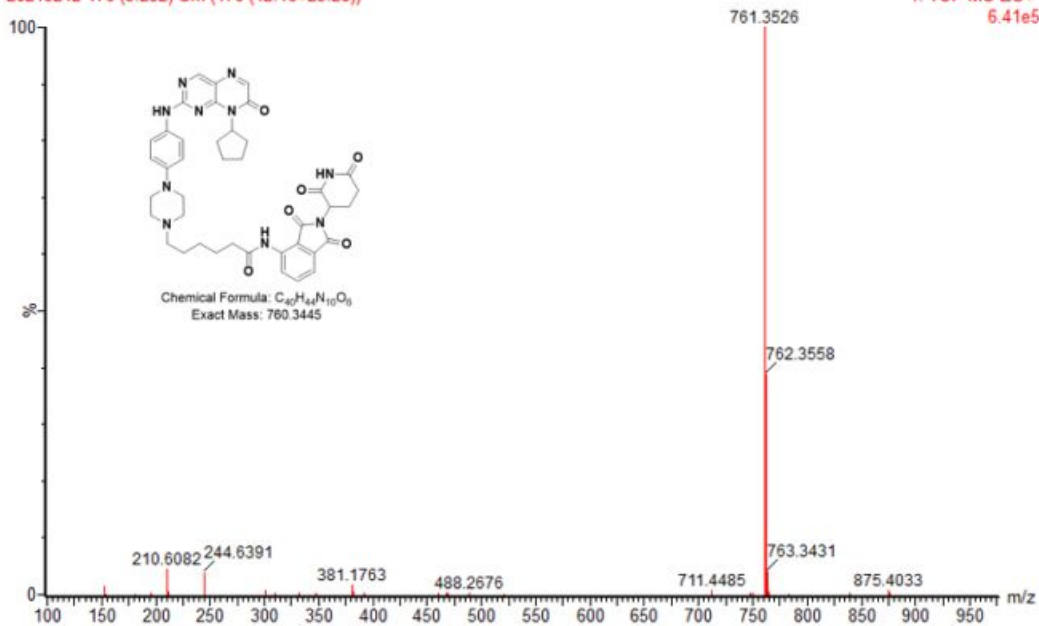


LQ-089-37

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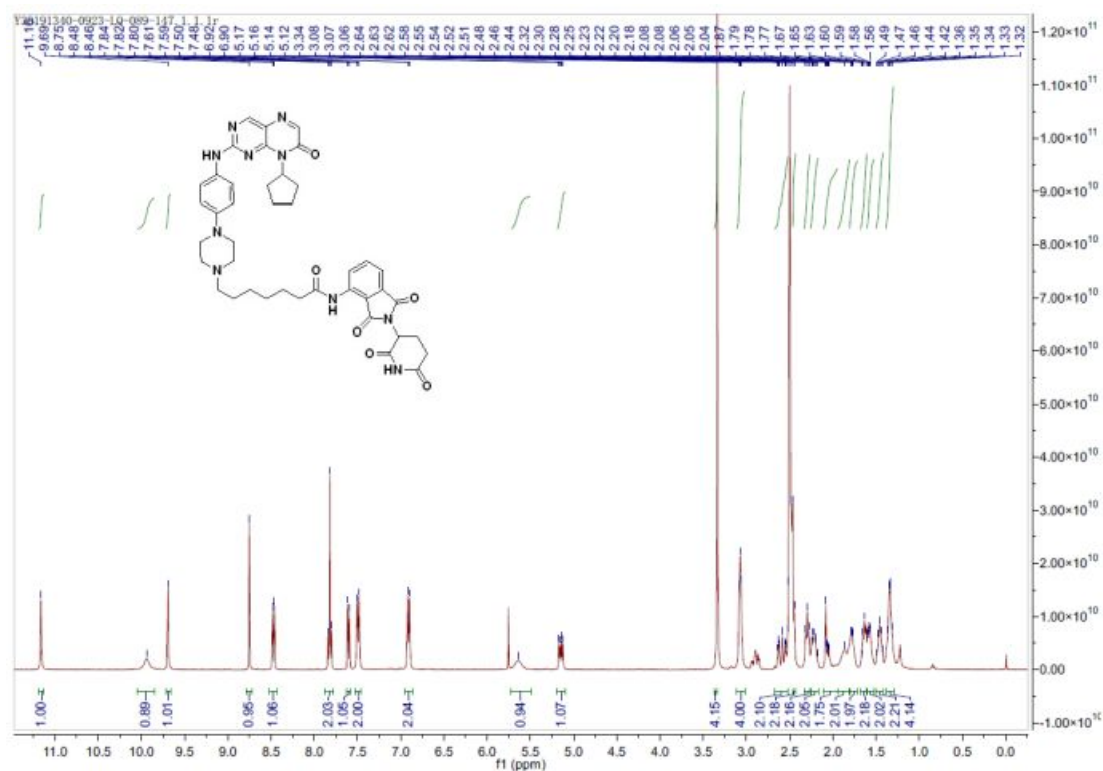
22-Sep-2021 13:26:13

20213212 176 (3.232) Cm (176-(12:16+23:28))

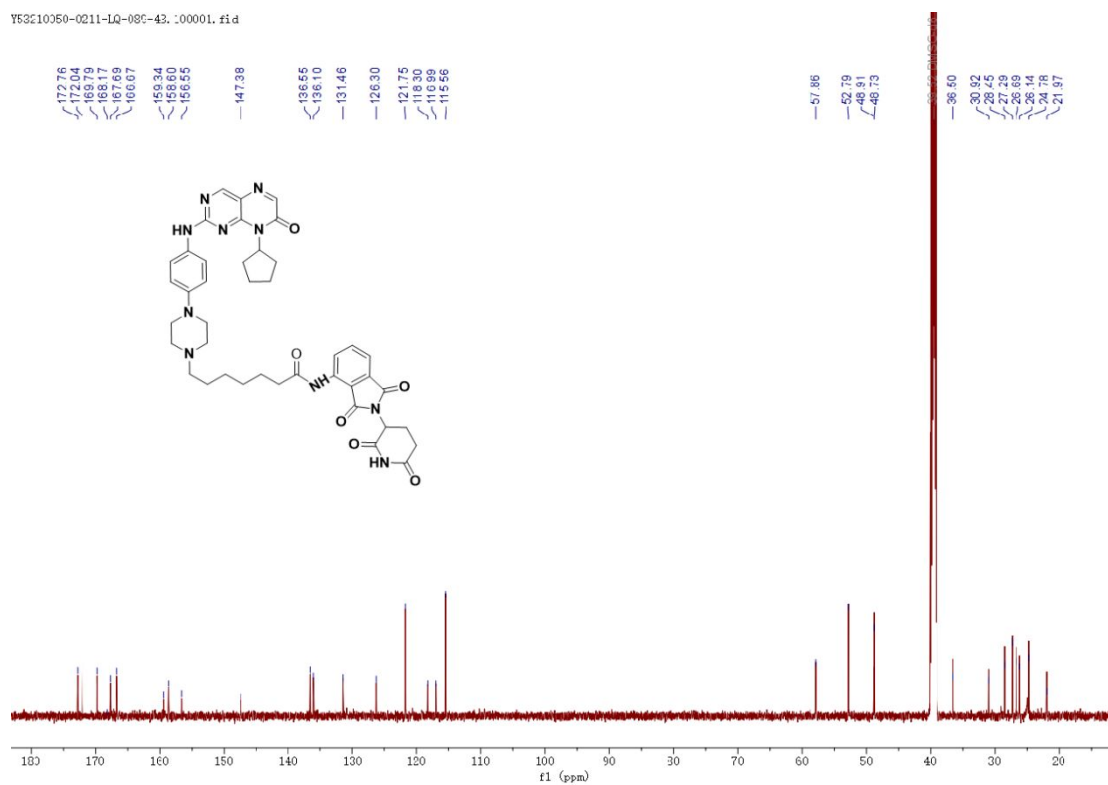
1: TOF MS ES+
6.41e5

No.	Retention (min)	Area	Area%
1	1.560	4913915	96.83
2	1.843	94406	1.86
3	2.088	66411	1.31

4. Spectra of compound 7d (^1H NMR, ^{13}C NMR, HRMS, HPLC)



Y5210050-0211-1Q-00C-43.100001.fid

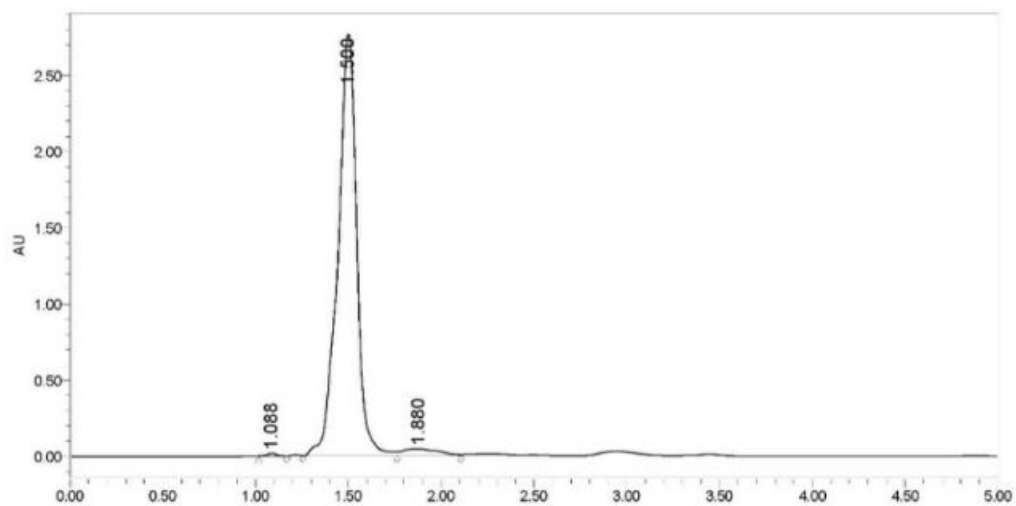
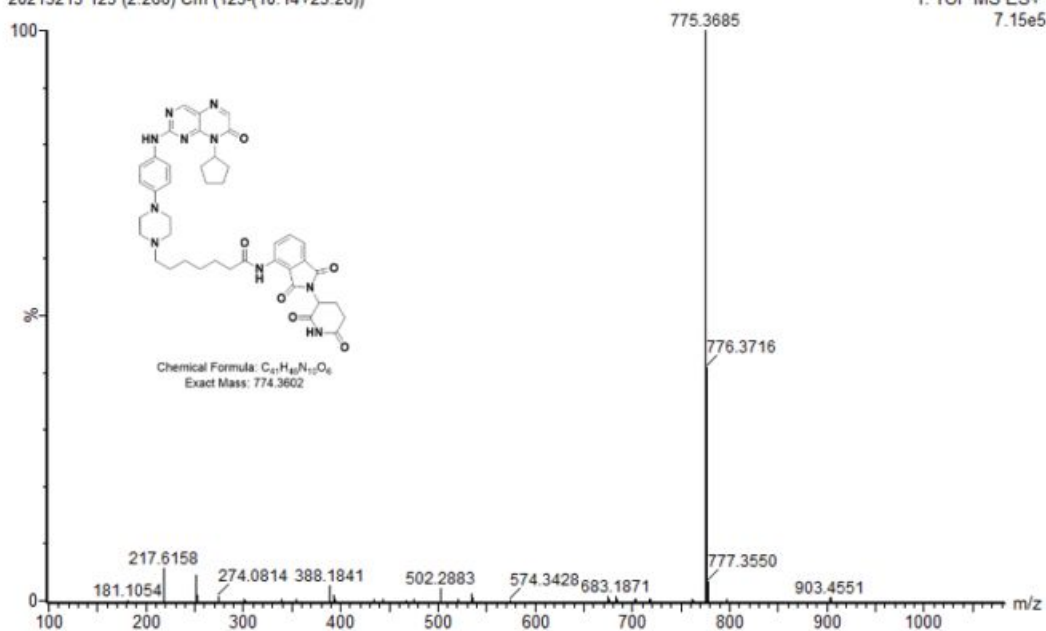


LQ-089-43

XEVO-G2TOF#NotSet

22-Sep-2021 13:33:46

20213213 123 (2.266) Cm (123-(10:14+23:26))

1: TOF MS ES+
7.15e5

No.	Retention (min)	Area	Area%
1	1.088	55386	0.27
2	1.500	19589513	96.67
3	1.880	618777	3.05

[illegible]

LQ-089-41

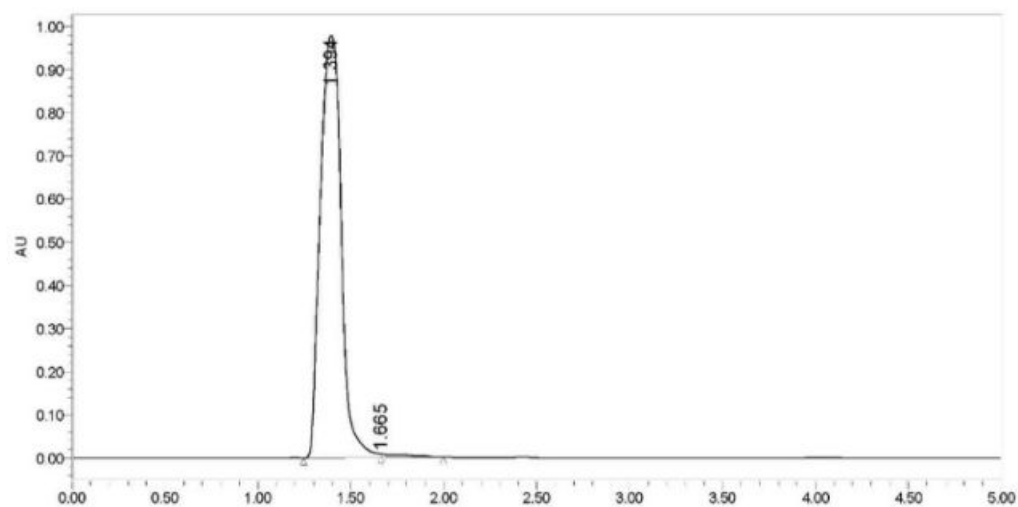
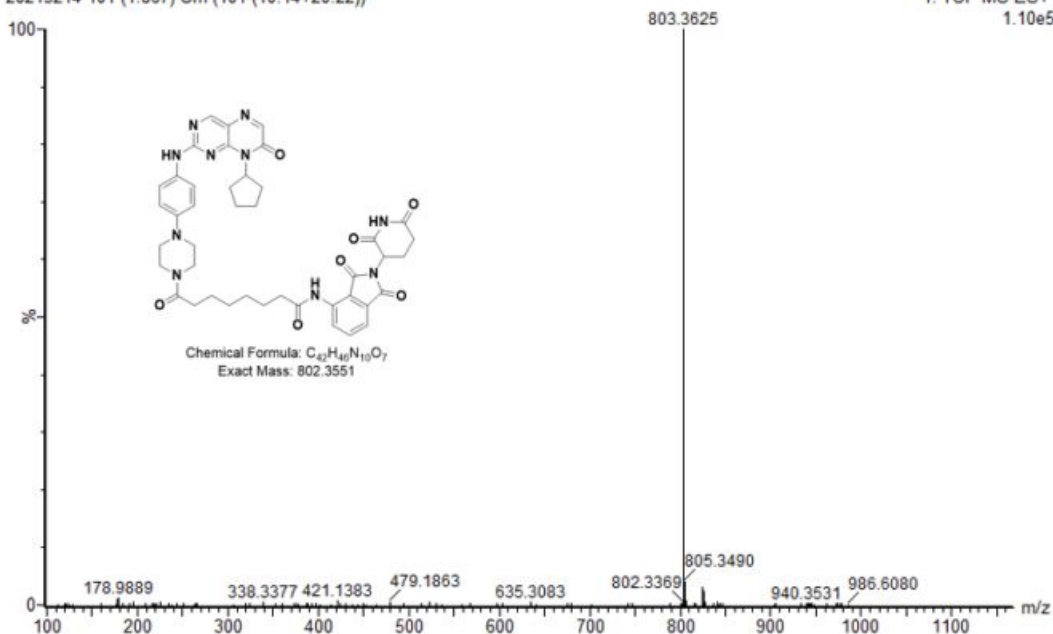
XEVO-G2TOF#NotSet

22-Sep-202113:42:31

20213214 101 (1.867) Cm (101-(10:14+20:22))

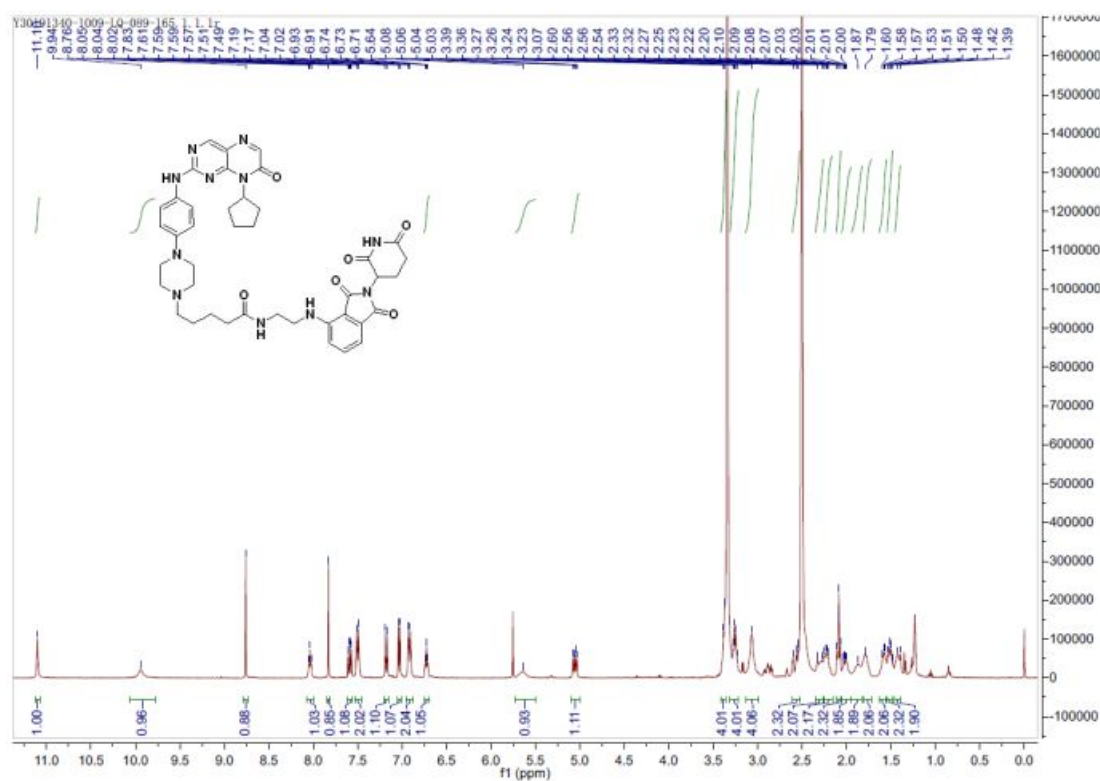
1: TOF MS ES+

1.10e5

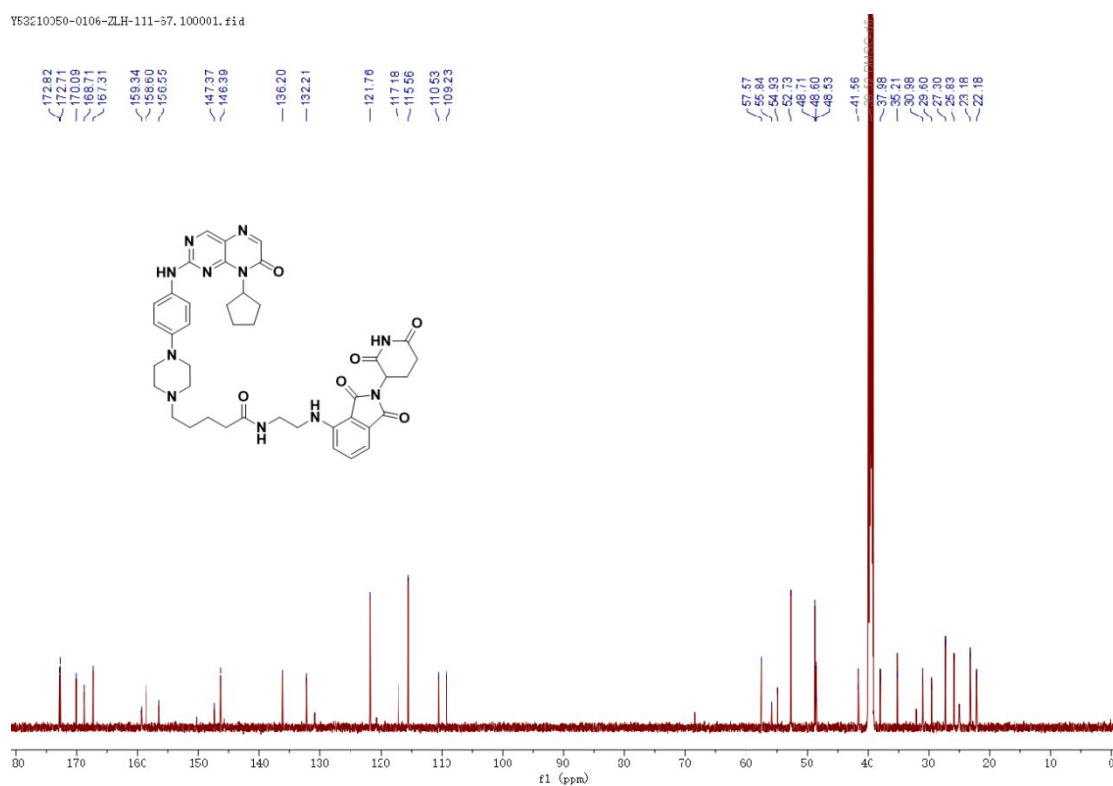


No.	Retention (min)	Area	Area%
1	1.394	7639316	98.89
2	1.665	85845	1.11

6. Spectra of compound 7f (¹H NMR, ¹³C NMR, HRMS, HPLC)



Y53210050-0106-ZLH-111-37.100001.fid



LQ-089-67

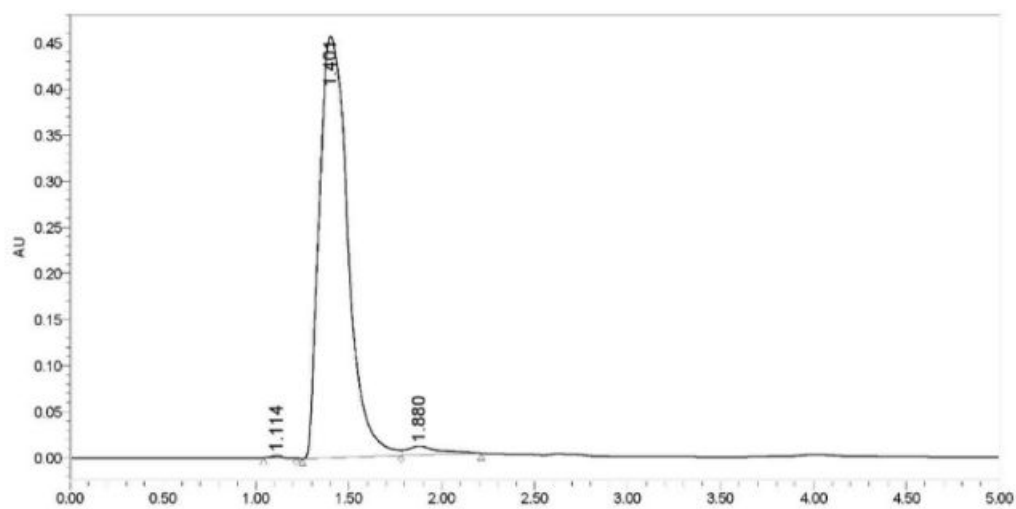
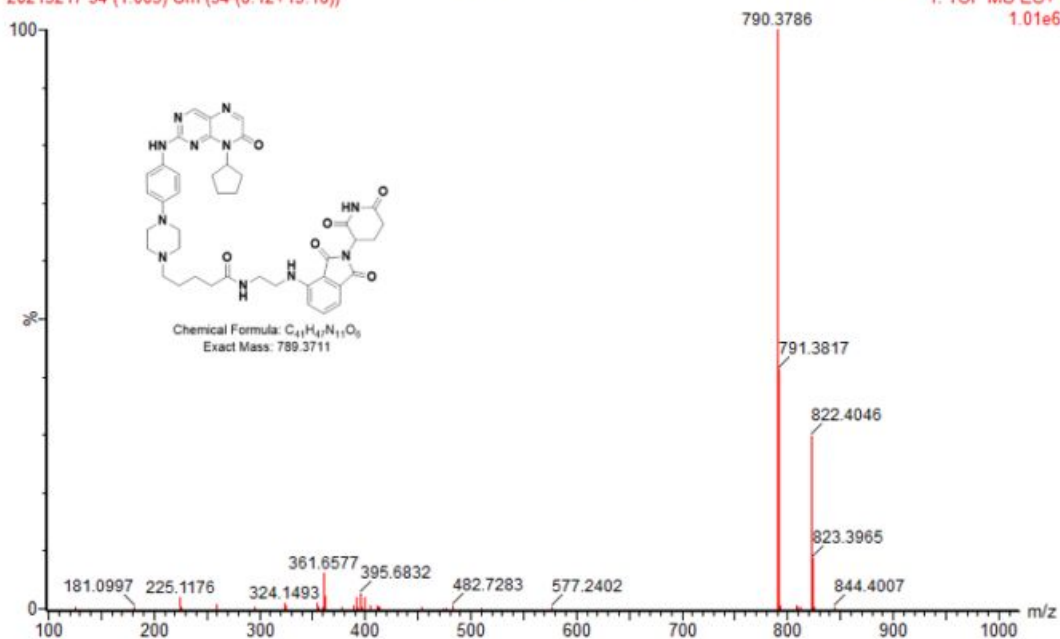
XEVO-G2TOF#NotSet

22-Sep-202114:01:17

20213217 54 (1.003) Cm (54-(8.12+15.18))

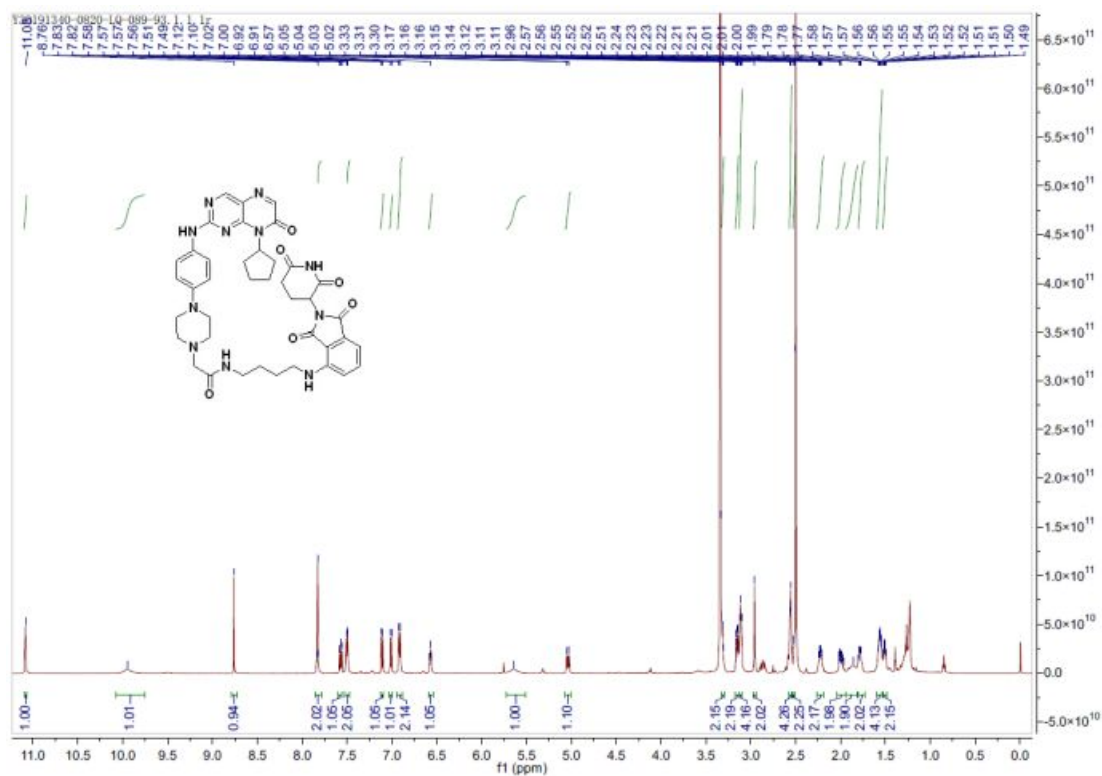
1: TOF MS ES+

1.01e6

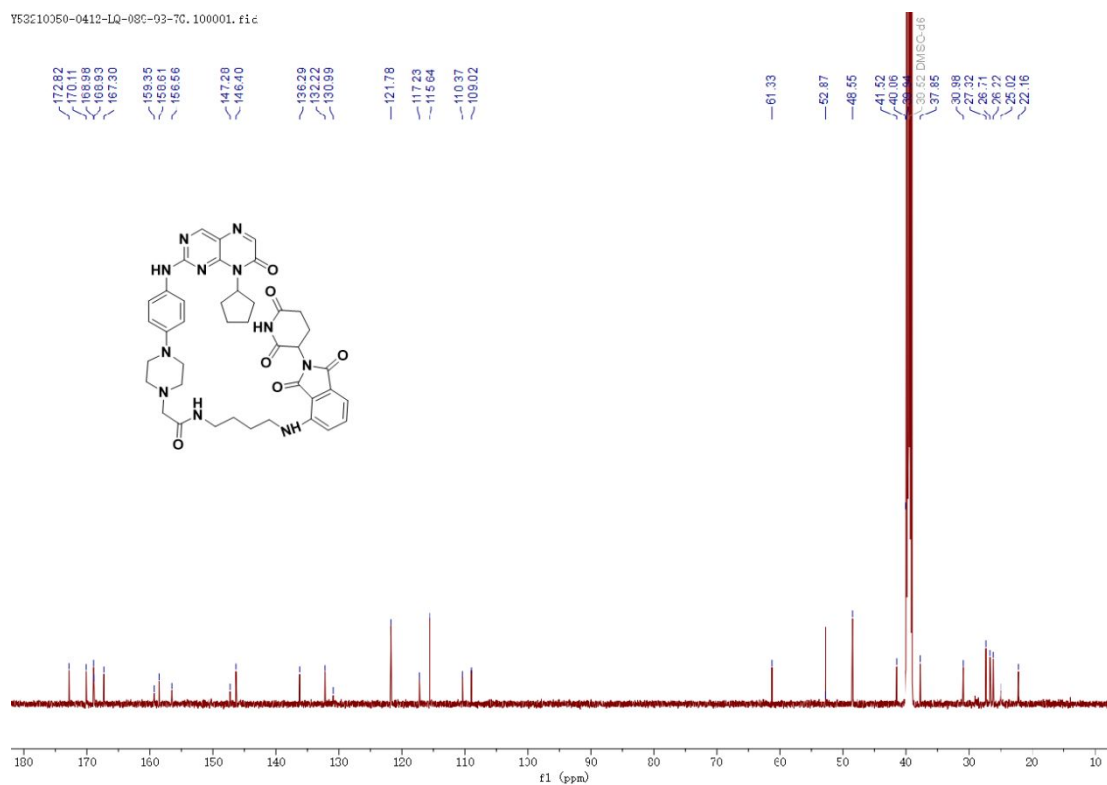


No.	Retention (min)	Area	Area%
1	1.114	11111	0.22
2	1.401	4869901	97.35
3	1.880	121240	2.42

7. Spectra of compound 7g (¹H NMR, ¹³C NMR, HRMS, HPLC)



Y53210050-0412-LQ-08C-03-70_100001.fic



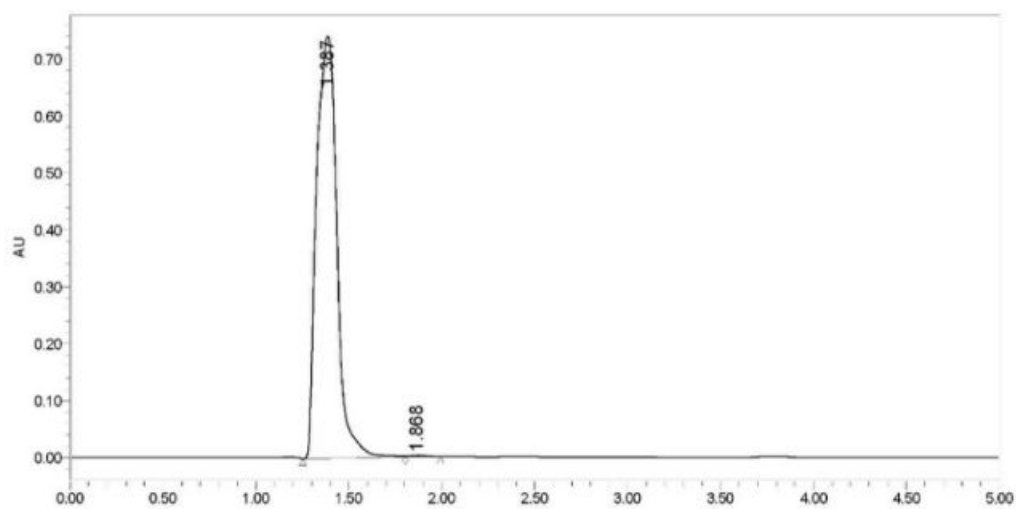
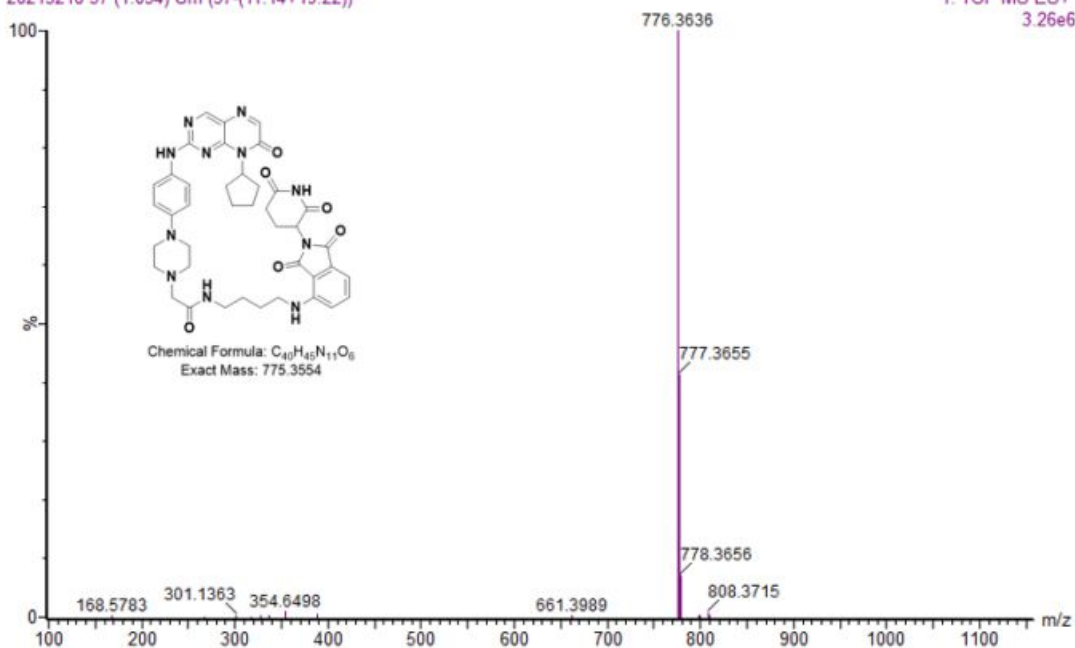
LQ-089-93

20213218 57 (1.054) Cm (57-(11:14+19.22))

XEVO-G2TOF#NotSet

22-Sep-202114:10:23

1: TOF MS ES+
3.26e6



No.	Retention (min)	Area	Area%
1	1.387	5637195	99.69
2	1.868	17674	0.31

Chemical structure of compound 103.1.1.1r is shown in the upper left. The structure is a complex molecule featuring a central piperazine ring substituted with a 1H-imidazo[4,5-b]pyridine-2-yl group, a 1-cyclopentyl-1H-imidazo[4,5-b]pyridine-2-yl group, and a 1-(4-oxo-1,2,3,4-tetrahydropyridin-2-yl)-1H-imidazo[4,5-b]pyridine-2-yl group.

¹H NMR spectrum (CDCl₃) of compound 103.1.1.1r. The x-axis represents the chemical shift in ppm (0.0 to 11.0), and the y-axis represents the intensity in arbitrary units (0 to 1,900,000). The spectrum shows several peaks, with integration values provided below the baseline.

Integration values (from left to right): 1.00, 0.03, 1.01, 0.93, 1.04, 1.00, 1.08, 0.96, 1.06, 2.01, 2.01, 1.00, 1.12, 2.09, 1.96, 6.32, 4.30, 2.20, 2.20, 1.07, 3.96, 2.08, 2.08, 2.11, 2.17.

Y5Z10J50-0412-LQ-08C-1J3-7H 100001.fid

The image displays the chemical structure of compound 10 and its corresponding ¹³C NMR spectrum. The chemical structure is a complex molecule featuring a central piperazine ring. One nitrogen of the piperazine is substituted with a 4-aminophenyl group. The other nitrogen is substituted with a 1-(cyclopentyl)-1H-imidazo[4,5-b]pyridin-5-yl group. The imidazopyridine system is further substituted with a 1-(cyclopentyl)-1H-imidazo[4,5-b]pyridin-5-yl group. The piperazine ring is also substituted with a 1-(cyclopentyl)-1H-imidazo[4,5-b]pyridin-5-yl group. The spectrum shows peaks from 10 to 60 ppm, with a major peak at 39.52 ppm. The peak list is as follows:

Chemical Shift (ppm)
172.77
171.87
169.90
168.25
166.87
159.31
158.80
156.55
136.50
136.09
131.47
126.35
121.76
119.34
117.24
115.68
57.55
51.78
48.17
48.51
39.52
38.96
30.93
29.40
27.39
26.35
25.02
24.11
23.05
21.88
19.39
19.20
13.50

LQ-089-103

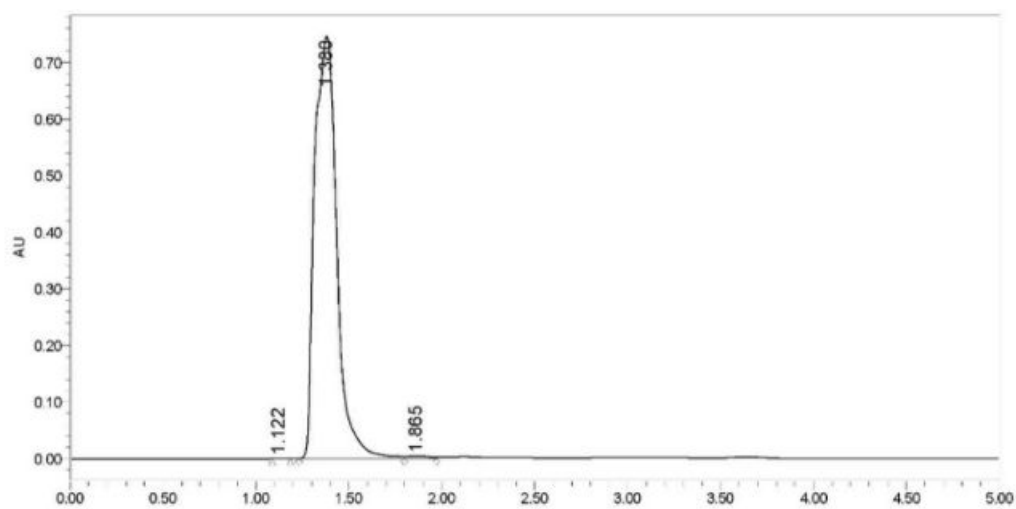
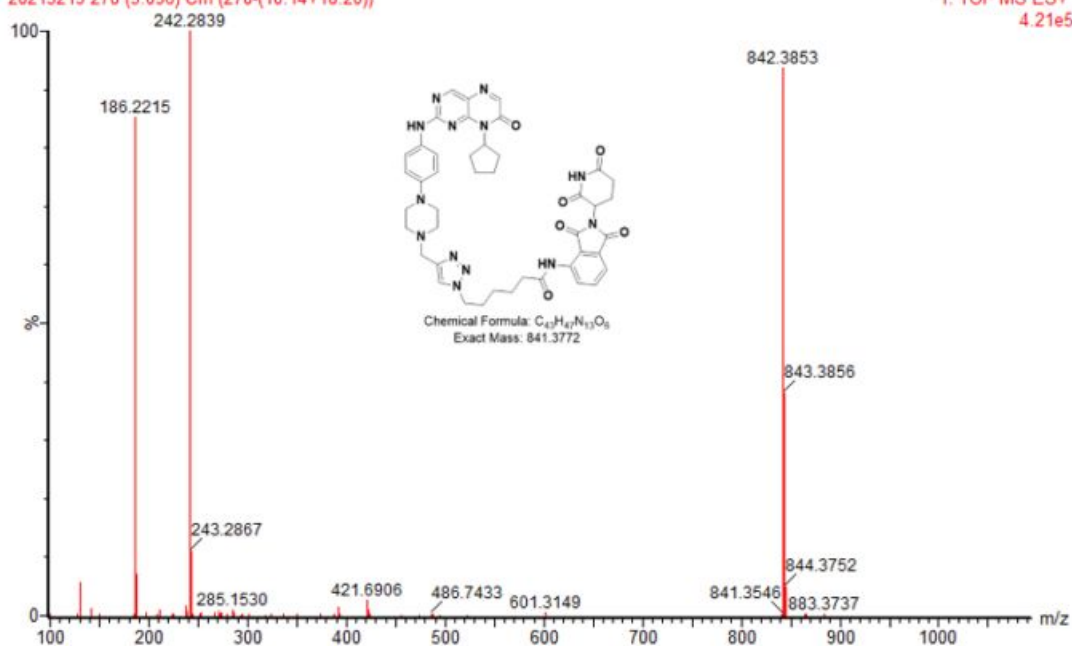
XEVO-G2TOF#NotSet

22-Sep-2021 14:17:10

20213219 278 (5.096) Cm (278-(10:14+18:20))

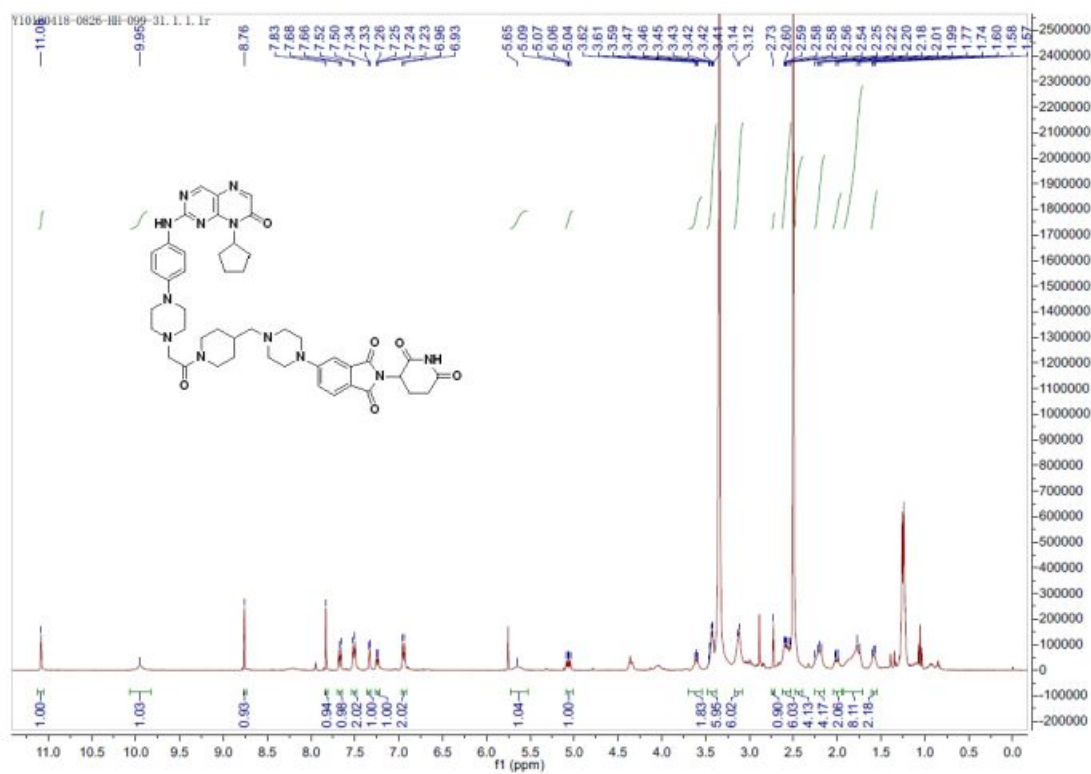
1: TOF MS ES+

4.21e5

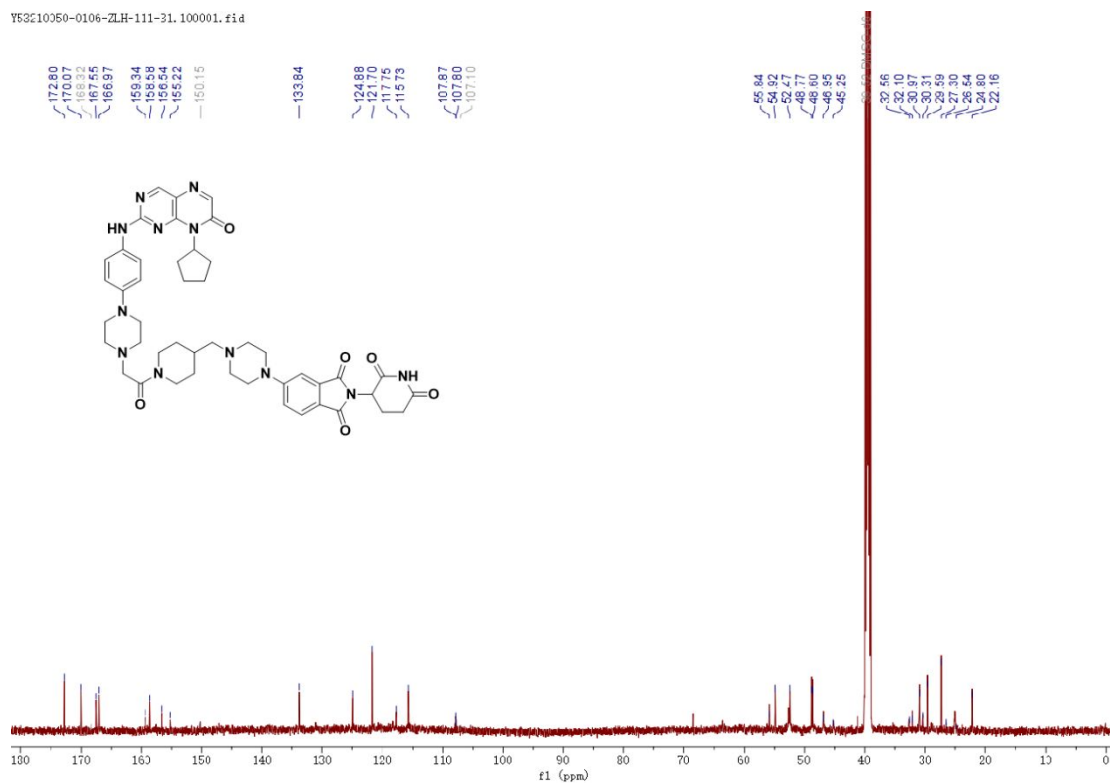


No.	Retention (min)	Area	Area%
1	1.122	1119	0.02
2	1.380	6163495	99.46
3	1.865	32111	0.52

9. Spectra of compound 7i (¹H NMR, ¹³C NMR, HRMS, HPLC)



Y53210050-0106-ZLH-111-31.100001.fid

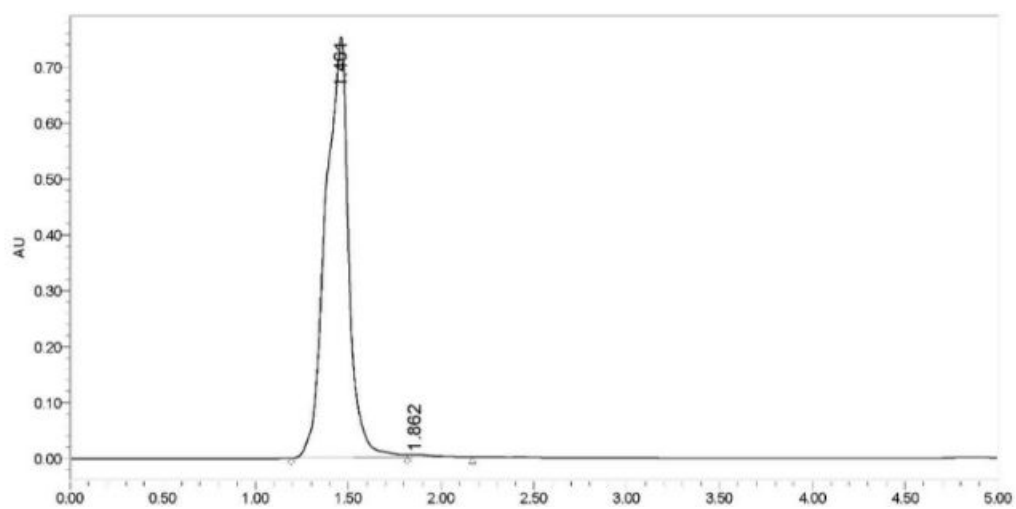
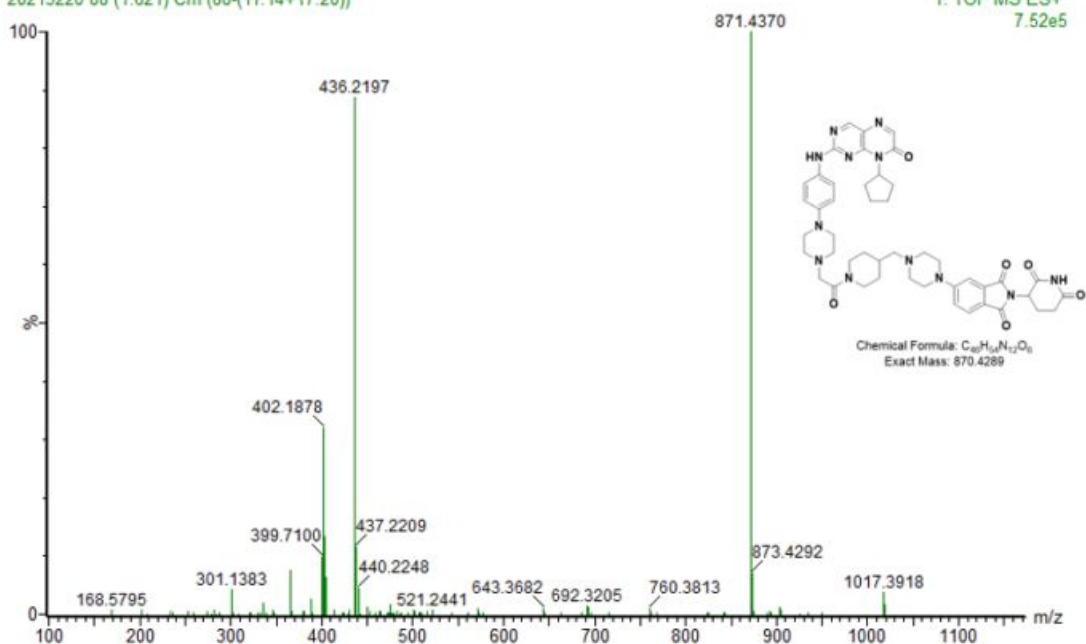


HH-099-31

XEVO-G2TOF#NotSet

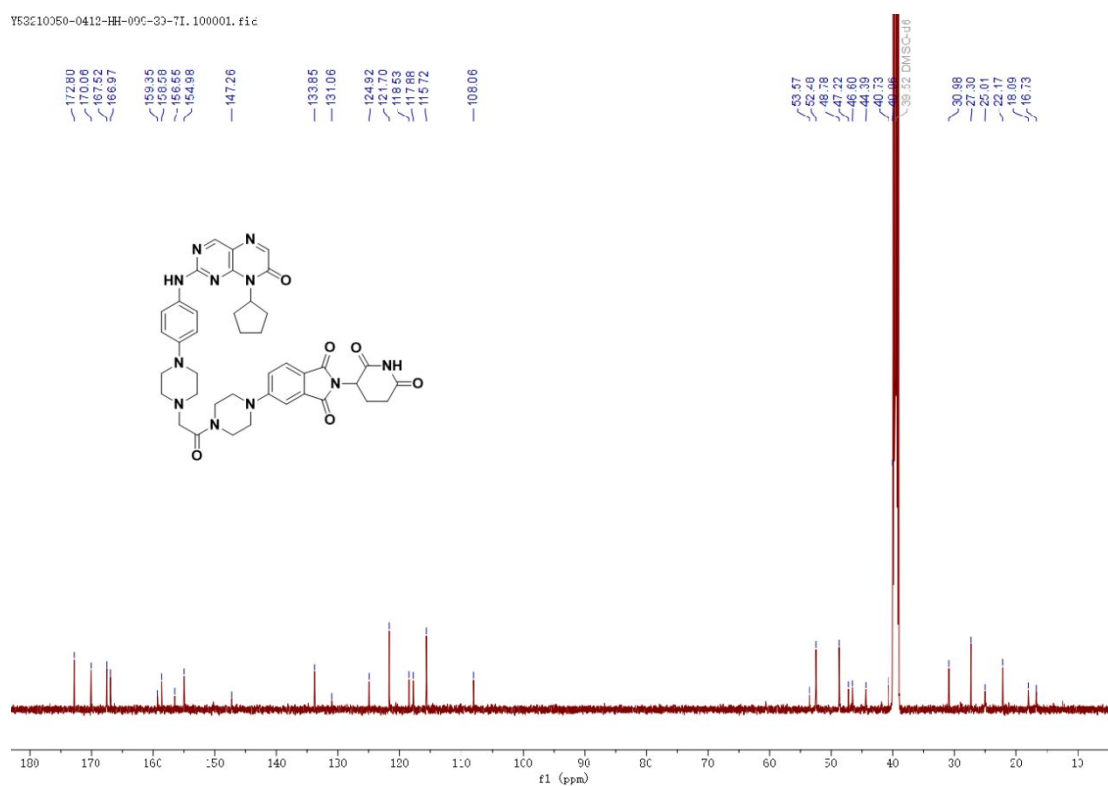
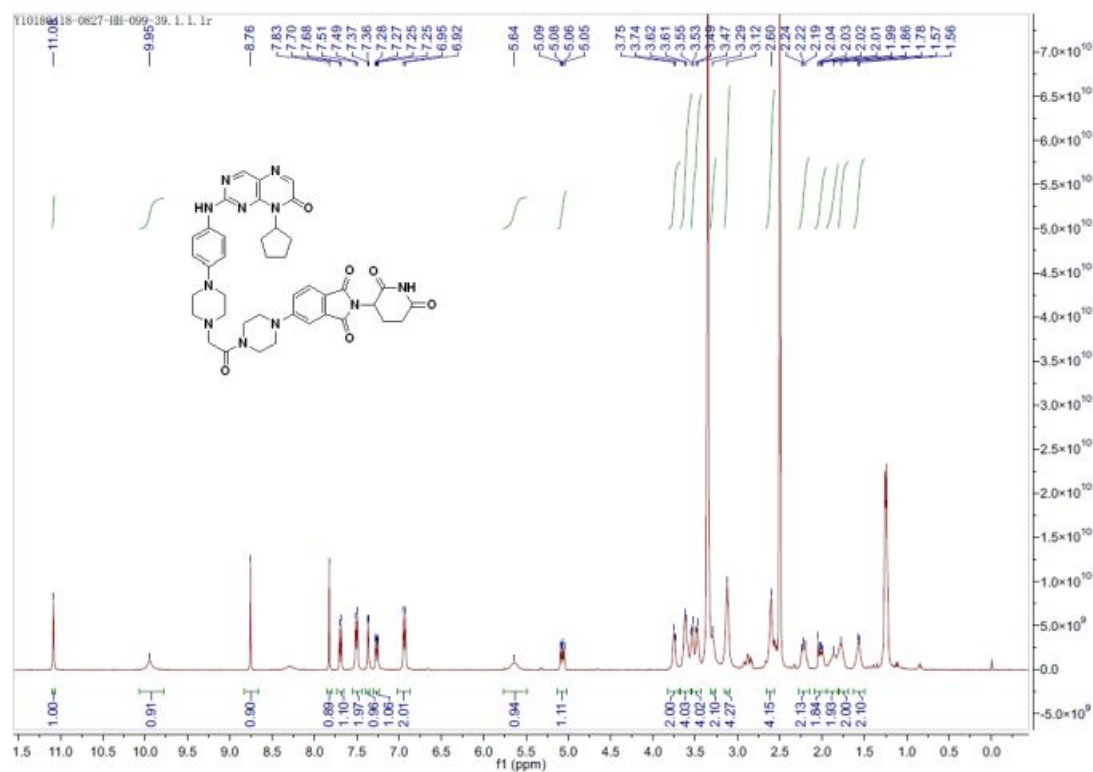
22-Sep-202114:27:18

20213220 88 (1.621) Cm (88-(11:14+17:20))

1: TOF MS ES+
7.52e5

No.	Retention (min)	Area	Area%
1	1.461	6516842	99.27
2	1.862	47869	0.73

10. Spectra of compound 7j (¹H NMR, ¹³C NMR, HRMS, HPLC)



HH-099-39

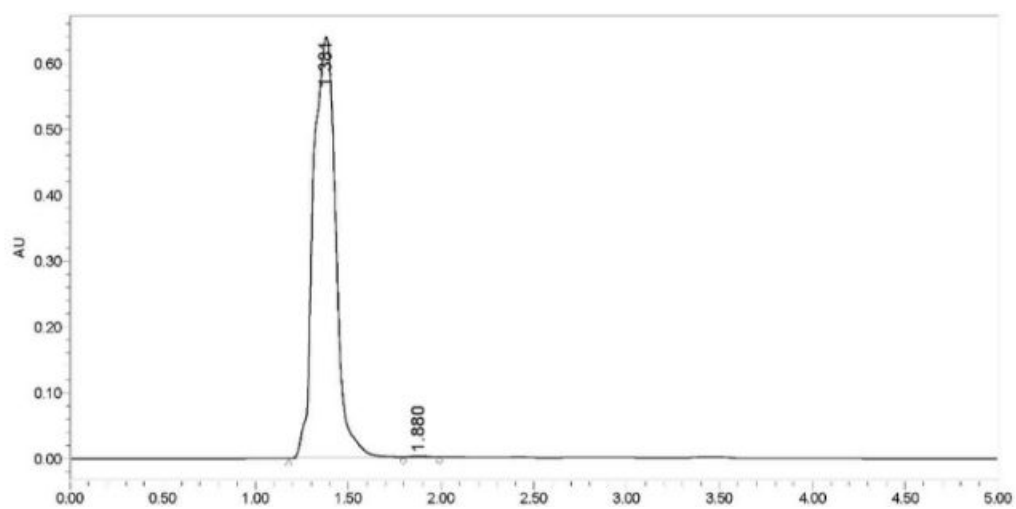
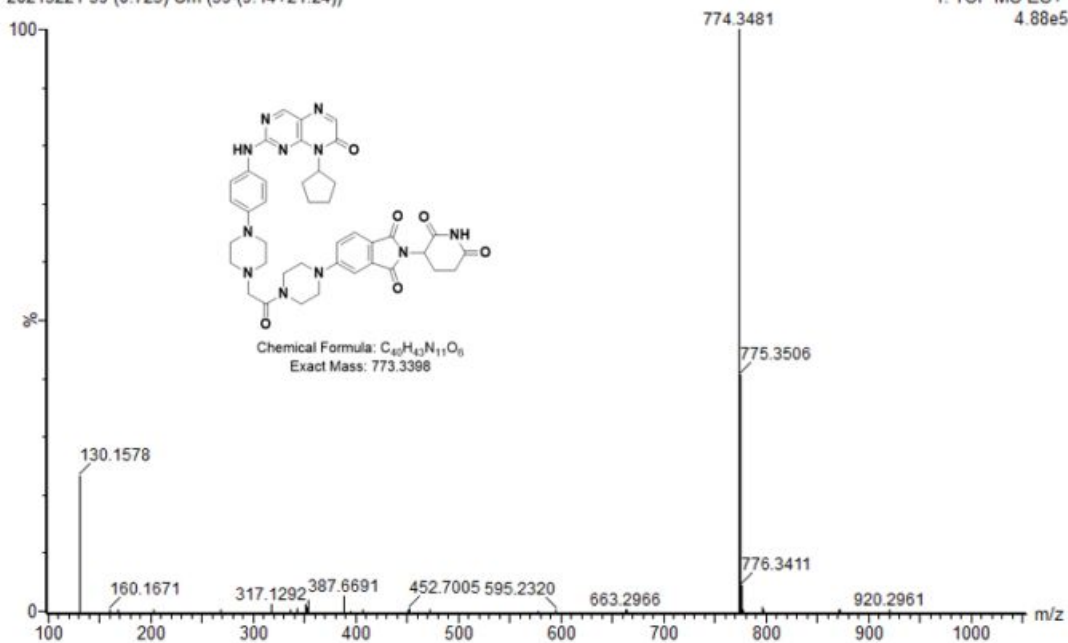
XEVO-G2TOF#NotSet

22-Sep-2021 14:36:49

20213221 39 (0.723) Cm (39-(9.14+21.24))

1: TOF MS ES+

4.88e5



No.	Retention (min)	Area	Area%
1	1.381	5336521	99.63
2	1.880	19680	0.37

4 Evaluation of the pharmacokinetic properties of 7f in rats.

The pharmacokinetic study of **7f** was evaluated in male Sprague–Dawley rats via intravenous (i.v., 1 mg/kg) and p.o. (10 mg/kg) administration.

Table S1. Pharmacokinetics parameters of compound 7f.

Compd	Route	C _{max} (ng/mL)	T _{1/2} (h)	AUC _(0-t) (ng/mL*h)	CL (mL/h/kg)	F (%)
7f	IV (1 mg/kg)	252.55±31.40	0.95±0.03	85.16±10.34	11312.91±1392.17	-
	PO (10 mg/kg)	NA	NA	NA	NA	-

C_{max} maximum observed plasma concentration, T_{1/2} elimination half-life, AUC area under the plasma concentration, Cl clearance, F bioavailability, NA means Not Available.